

Glia Cell Dynamics in the Zebrafish Optic Tectum

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List of Abbreviations & Definitions

<i>atoh 7</i>	Atonal homologue 7 - retinal ganglion cell differentiation transcription factor
CMZ	Ciliary marginal zone
CNS	Central nervous system
dpf	Days post-fertilization
E3	Embryo medium three
GFP	Green Fluorescent Protein
hpf	Hours post-fertilization
<i>lak</i>	<i>lakritz</i> – mutant carrying a non-sense allele in <i>atoh5</i> gene, causing loss of RGCs
LGN	Lateral geniculate nucleus
LSM	Light-sheet microscope
OPC	Oligodendrocyte Precursor Cell
OT	Optic Tectum
process	Cell extension, appendage, or protrusion (axon, dendrite, etc.)
RGC	Retinal ganglion cell
SC	Superior Colliculus
Teg	Tegmentum
Tg	Transgenic
TSc	Torus semicircularis
WT	Wild Type (as opposed to experimental or mutant)

Table of Contents

Introduction.....	1
Zebrafish Model.....	1
Zebrafish visual system development	2
Visual System Overview	2
Early Development of the Central Nervous System (CNS)	3
Development of the Eye	5
<i>lakritz</i> Mutation	9
Development and Organization of the Optic Tectum	9
Development and Refinement of the Retinotopic Map	11
Retinotopic Map Refinement.....	14
Adult Neurogenesis.....	14
Oligodendrocytes	17
Development	17
OPCs	17
Microglia	19
Light-Sheet Fluorescence Microscopy	20
Present Study.....	22
Methods	25
Transgenic lines	25
Zebrafish Husbandry	25
Rearing Embryo/Larvae	26
Imaging	26
File Naming Conventions.....	26
Prep	27
Mounting	28
Laser Calibration.....	29
Image Acquisition	30

Tradeoffs.....	31
Image Processing	32
Maximum Intensity Timelapse	33
Luxendo Image Processor	33
Image Analysis	36
Manual Spot Creation	37
Surface Creation	37
Masking out a channel	39
Microglia Analysis.....	39
Results	43
Nerve Growth and Nerve Coordinated OPC migration	43
Optic Nerve Growth Patterns	45
The <i>nkx2.2a</i> Reporter Labeling Specificity	47
Early <i>lak</i> and WT Qualitative OPC Comparison	52
WT and <i>lak</i> mutants have differing OPC cell count and morphology	57
Lightsheet live timelapses show a variety of dynamic microglia behaviors	59
Quantification of Microglia between WT and <i>lak</i>	60
Discussion	65
RGC growth and OPC migration.....	65
RGC associated OPCs	66
Questions from <i>nkx2.2a</i> labeling specificity.....	67
Microglia.....	67
Future directions.....	68
Appendix A: Additional Protocols.....	71
Embryo Micro-Injections.....	71
Eucleations	71
Appendix B: Additional Figures.....	73
Microglia show dynamic interactions with cells undergoing apoptosis	73
Bibliography.....	75

List of Figures

Figure 1: Steps of Zebrafish Neurulation	4
Figure 2: Formation of the optic cup following neurulation	7
Figure 3: Optic nerve growth and defasciculation	8
Figure 4: Topographic mapping from RGCs in the retina to the neuropil of the Optic Tectum.	12
Figure 5: Proliferative zones in the zebrafish visual system.....	16
Figure 6: OPCs form synaptic contacts with neurons.	18
Figure 7: Cell lineages of hematopoietic and CNS glia precursors.....	19
Figure 8: Dual-illumination/dual objective light-sheet microscopy.....	22
Figure 9: Demonstration of a surface macrophage with reduced tissue depth- related aberration.....	41
Figure 10: 2-3 dpf time-course of optic nerve growth and OPC migration.....	44
Figure 11: Defasciculation of the optic nerve in relation to the pre-optic tract	46
Figure 12: Differences in the range of cells and structures expressing <i>nkx2.2a:GFP</i>	51
Figure 13: Comparison of WT and <i>lakritz</i> OPCs at 3 dpf	53
Figure 14: Comparison of WT and <i>lakritz</i> OPCs at 5 dpf	57
Figure 15: Quantification of WT and <i>lakritz</i> OPCs at 5, 7, and 9 dpf.....	59
Figure 16: Microglia Apoptosis, Phagocytosis, and Mitosis	60
Figure 17: Comparison between WT and <i>lakritz</i> microglia at 5 dpf.....	63
Figure 18: Microglia colocalization with apoptotic cells	73
Figure 19: Microglia colocalization with Neutral Red stained lysosomes.....	74

Abstract

The development of the brain requires countless modes of interaction between cells. Sensory systems send axon projections that connect to the brain (innervation), allowing sensory modalities to be processed. This sensory innervation also influences the development of the target regions. The zebrafish visual system is a good model to study this innervation-dependent development. Neurons in the retina called retinal ganglion cells (RGCs) transmit visual information through their axon projections, which make up the optic nerve, into the optic tectum (OT) in the dorsal midbrain, which results in a topographic map of visual information. The connection between the optic nerve and the OT is important for the survival and growth of cells in the OT. However, many specifics of how this is mediated, as well as other innervation-dependent cellular dynamics, remains to be studied.

In this thesis, I used time-lapse light sheet microscopy to study innervation-dependent cell dynamics of microglia and oligodendrocyte precursor cells (OPCs), which are both glial cells (non-neuronal CNS cells) present in the OT that are known to facilitate neuronal remodeling. I utilized exploratory methods to observe OPC migration, localization, morphologies, and interactions with RGC axon terminals in WT larvae, and to additionally visualize microglia behavioral dynamics in the OT.

Introduction

Developmental Neuroscience is the field concerned with how the central nervous system (CNS) comes to be through cellular genesis, patterning, differentiation, and maturation. Much of the theoretical focus in developmental neuroscience research has to do with the human brain and how its development works or how normal development can go wrong and lead to deadly or debilitating disorders. However, most types of neuroscience research utilize model organisms because research on humans is often infeasible due to ethics, difficult or impossible methodology, and associated expenses. The most common model organisms are rats and mice, which, as mammals, are very analogous to humans in their biology while still being small and relatively easy to work with and develop quantitative methodology for. For many questions in neuroscience, however, especially many of those regarding development, mammalian models aren't preferable.

Zebrafish Model

Zebrafish (*Danio rerio*) are a small tropical teleost fish that can serve as a fantastic model organism. They have a variety of traits and advantages that have led to their success in the advancement of countless scientific fields since the start of their adoption into use as a model organism in the 1970s. For questions in developmental biology, they are very useful because their embryos and larvae are externally developing and mostly transparent. They also develop fast, being able to perform a variety of behaviors, including hunting and predator evasion by 5 days post-fertilization (dpf). These factors make them ideal for imaging biological processes occurring in live organisms. It is relatively easy and less

expensive than many other model organisms to take care of a colony of adult zebrafish. And these adults can then be bred at any time of year, multiple times a month, to produce up to hundreds of eggs in one clutch. Additionally, the zebrafish genome is fully sequenced, and very well studied, leading to a variety of genetic tools being developed for use with zebrafish, including robust transgenic lines that can express fluorescent proteins associated with specific genes in order to fluorescently label specific cell types and/or structures which can then be imaged using fluorescent microscopy in live developing embryos and larvae (Glass & Dahm, 2004).

Zebrafish visual system development

Visual System Overview

The flow of visual information is relatively conserved between vertebrates, especially in the eye. Vision involves transducing light into a sensory signal that can be processed in the brain. The first step in this process occurs when light is focused by the lens onto the retina and then interacts with proteins known as rhodopsins in the membranes of photoreceptor cells. This starts the phototransduction cascade by causing the photoreceptor cells to hyperpolarize and ultimately activate retinal ganglion cells (RGCs). RGCs have long axons that make up the optic nerve and project into several regions of the brain. In mammals, the primary region that receives optic innervation is the lateral geniculate nucleus (LGN), which then relays information to the primary visual cortex. Additional retinorecipient areas are the pretectal areas involved in reflexive visual responses, and the superior colliculus (SC). The zebrafish analogue to the SC is the optic tectum (OT), which is the primary retinorecipient area of the brain. Both the SC and OT are responsible for forming a topographic map of sensory information they receive from the retina and coordinating space

specific visual information to inform behavior directed towards the corresponding space in the world.

Early Development of the Central Nervous System (CNS)

Some of the earliest stages of development involve movement and separation of germ layers and specification of broad cell fates. The zebrafish embryo undergoes involution of the hypoblast, which will later be separated into the endoderm and mesoderm. Just after 5 hours post fertilization (hpf), the germ layers are at 50% epiboly and the germ ring forms in two distinguishable layers at a roughly uniform thickness. At 6 hpf, there is a slight thickening in the epiblast and hypoblast indicating the embryonic shield, which marks the eventual dorsal side of the embryo (Warga & Nüsslein-Volhard, 1999), (Kimmel et al., 1995).

Later involuting cells from the hypoblast generally form the mesoderm. This presumptive mesoderm comes in contact with the dorsal ectoderm during early gastrulation. The dorsal mesoderm secretes signaling factors in a coordinated manner to the overlying dorsal ectoderm that induce specification of the neuroectoderm and establish basic anterior-posterior patterning. (Doniach & Musci, 1995). The most anterior segment of the neuroectoderm goes on to become the forebrain (prosencephalon), followed posteriorly by the midbrain (mesencephalon), and then the hindbrain (rhombencephalon) while the most posterior section becomes the spinal cord. Once the neuroectoderm is specified, it begins to form the neural plate which becomes a pseudostratified epithelial tissue that folds into a neural tube. Zebrafish neural tube formation undergoes a process different from many other vertebrates including mammals, which form a hollow neural tube immediately by rolling the neural plate inwards to join the dorsal midline. The teleost neural plate converges to first form the neural keel, then, the dorsal midline is fused to make a solid neural rod. Then the neural rod is hollowed by symmetric cell divisions in the medial neural rod, and the

neurocoele is formed, a hollow cavity which will go on to form the ventricles in the brain. Thus, the neural tube is formed (Figure 1), which allows the structural organization that allows primary neurogenesis to start. (Schmidt et al., 2013).

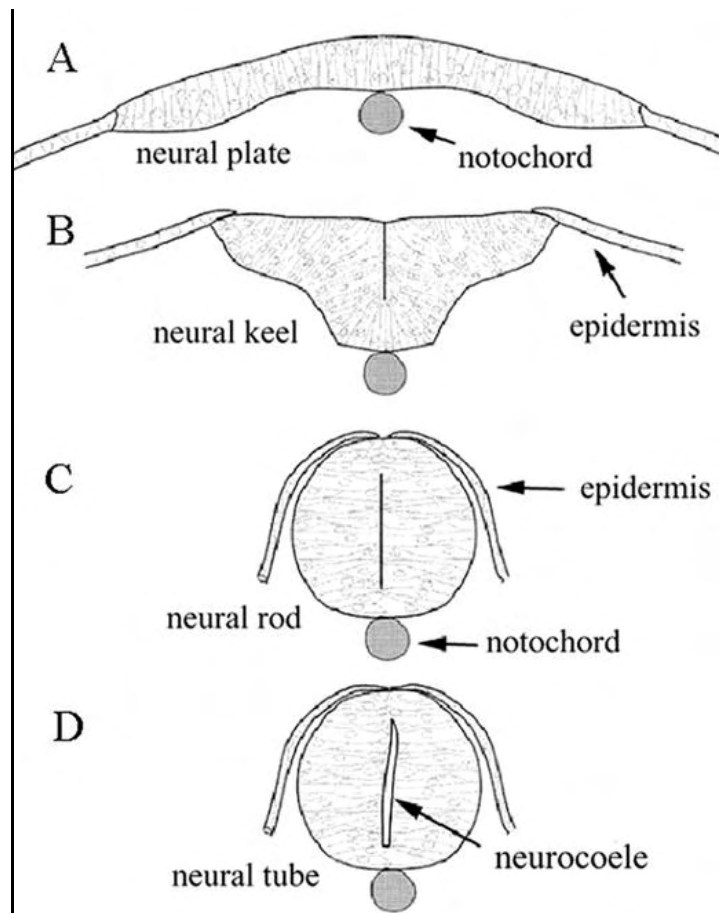


Figure 1: Steps of Zebrafish Neurulation

Diagram showing four key steps of neural tube formation in early CNS development. The views are transverse horizontal sections of an embryo mounted dorsal surface up. Figure reprinted from (Geldmacher-Voss et al., 2003).

Neuroepithelial cells in the neural tube have apical-basal polarity, which is important for defining the movement, structure, and fate of early neural cell populations. At this stage, there are many asymmetrically dividing cells, where the basal daughter stays in the apical progenitor pool but the apical daughter

becomes a neuronal progenitor. The neuronal progenitors then migrate from the apical zone to the basal zone of the neural tube, where they can differentiate into mature neurons (Dong et al., 2012).

Development of the Eye

Before the formation of the neural tube, the patterning of the neural plate must be specified. There are many signaling pathways likely at play here including Wnt/ β -catenin. The anterior-posterior patterning of the neuroectoderm is established in early gastrulation and leads to specification of the forebrain in the anterior neural plate. This is then further specified into several regions including a single field of eye precursor cells that express eye specification transcription factors. This eye field is maintained through local upregulation of Wnt1, which truncates the eye field and other rostral forebrain sections and preserves their patterning.

As the neural tube forms during neurulation, the eye field precursor cells must migrate to the lateral edges of the neural tube and then evaginate to form two optic vesicles on the left and right side. A certain level of epithelial remodeling is likely necessary for evagination, but in zebrafish, only some of the eye field cells are epithelialized at the start of optic evagination, and the rest migrate freely and then continue to epithelialize on the margin of the eye field as the optic vesicles form (Bazin-Lopez et al., 2015). Additionally, throughout the formation of the optic vesicles, there is a gradual 'flow' of cells from the neural tube core with early migrating cells ending up in distal regions like the optic cup, and later migrating cells ending up in medial regions like the optic stalk (Ivanovitch et al., 2013). In this early stage of eye development, the cells are not specified to different fates. The morphogenic flow of intercalating cells leads to those cells encountering several domains of the forming eye and brain, which may deliver key cues for regional patterning in the eye such that the trajectory of an intercalating cell may indicate its eventual fate (Bazin-Lopez et al., 2015).

The optic vesicle is patterned according to a proximal/distal axis. Proximal cells express the transcription factors *pax2* and *vax1*, which lead the cells to develop into the optic stalk. Distal cells express *pax6* and develop into the optic cup. In distal cells, the interaction between the protruding optic vesicle and the outer lying surface ectoderm is the main thing that drives the fate specification of the neural retina cells, lens ectoderm, and the retinal pigment epithelium (RPE), which all make up the optic cup. In a period from 16-24 hpf, the distal cells of the optic vesicle invaginate to form the optic cup there with an inner layer made up of the neural retina, specified by direct contact with the outer ectoderm, and an outer layer of RPE, which has close interaction with the extraocular mesenchyme. The ciliary margin is formed at the peripheral edge of the optic cup, or the transition point between the RPE and neural retina. This goes on to form the ciliary body and iris and in zebrafish a small group of cells are maintained to make the ciliary marginal zone, which is a retinal stem cell niche. The lens is formed when neural retina-specified cells send signals to the outer ectoderm layer that induce invagination into the optic cup. The invagination process starts at the ventral edge, and at the end, the optic fissure or choroid fissure remains as a narrow opening on the ventral side of the optic cup, which is important to allow the growth of the optic nerve and vasculature in the eye (Fuhrmann, 2008).

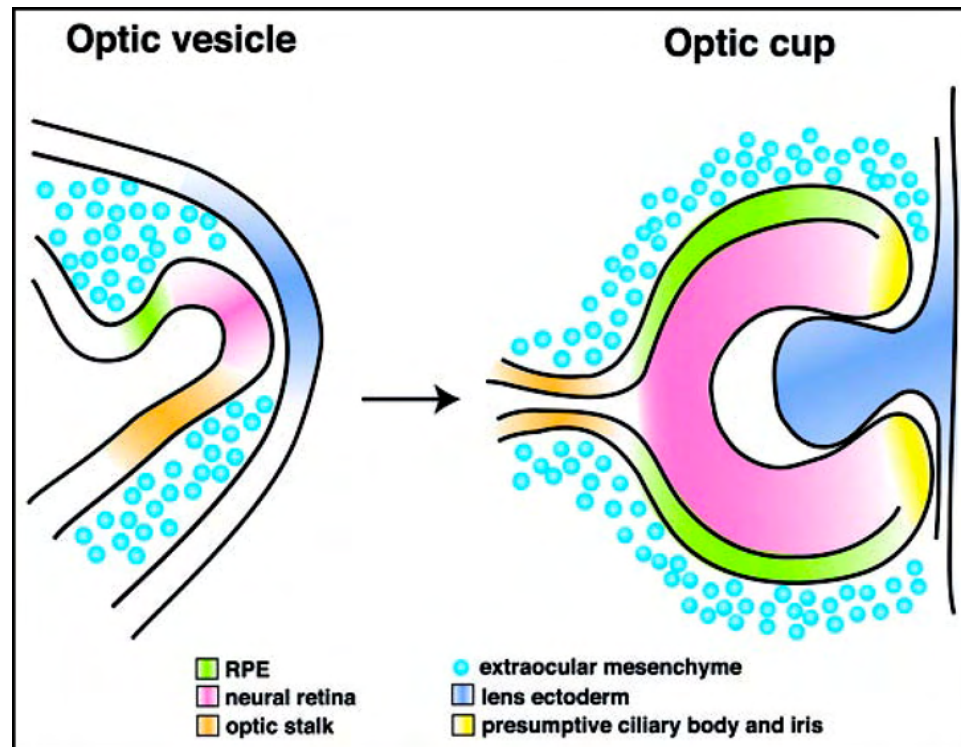


Figure 2: Formation of the optic cup following neurulation

Diagram showing the location of populations of regions/cell types as the retina transitions from an optic vesicle to an optic cup. Abbreviations: RPE (retinal pigment epithelium). Reprinted from (Fuhrmann, 2008)

The neural retina is patterned along the nasal/temporal, dorsal/ventral, and proximal/distal axes through a combination of signaling pathways. These pathways are also important for the differentiation of neurons and growth of the retina (Valdivia et al., 2016)

Differentiation in the retina occurs slowly during the formation of the optic vesicle, but then increases rapidly at around 24hpf in a wave of proliferation that starts with a ventronasal population of central retinal progenitor cells (RPCs) and then spreads outwardly to the periphery and proceeds circumferentially around the retina until the entire central retina is filled by ~48 hpf. From this point on, neurogenesis occurs in the ciliary marginal zone (CMZ) in the periphery of the retina. After the primary wave of proliferation, RPCs exit the cell cycle and differentiate into retinal ganglion cells (RGCs), interneurons, rod and cone

photoreceptors, and Müller glia (Valdivia et al., 2016). The wave of proliferation is accompanied by a wave of signaling molecules that control cell cycle timing and differentiation into the variety of retinal tissues. The transcription factor *atoh7* is responsible for conferring differentiation of RPCs into RGCs (Covello et al., 2020). Once RGCs are differentiated, they send out axon projections which bundle to make up the optic nerve that exits through the choroid fissure and migrates along the optic stalk toward the midline. After crossing the optic chiasm, the optic nerve turns dorsally and enters into the optic tectum (OT). Once in the OT, the bundle of axons separates (defasciculation) and axons innervate their individual targets in the OT neuropil region (Figure 3). In Zebrafish, the optic nerve exclusively projects contralaterally to the OT, meaning they have monocular vision. However, they have binocular integration thanks to neurons in the OT that project inter-tectally and interact with ipsilateral neurons in the opposite side of the OT. This allows them to more effectively hunt when prey are positioned in their binocular strike zone (Gebhardt et al., 2019)

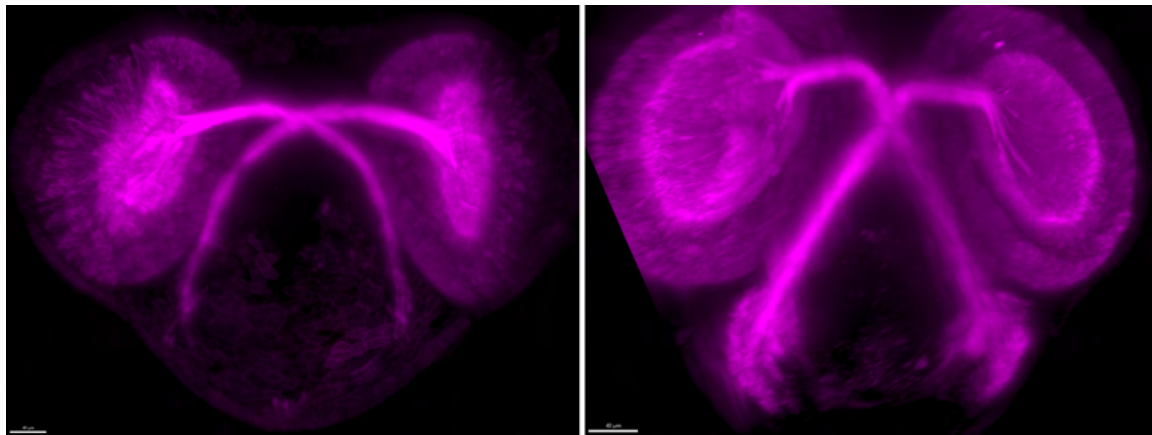


Figure 3: Optic nerve growth and defasciculation

Representative maximum intensity projection images captured from a ventral-caudal view showing the eyes and optic nerves. RGCs are labeled in magenta with Tg(*atoh7*:RFP). Left image is from a ~2 dpf larva. Right image is from a 3 dpf larva.

RGCs innervate the OT in a coordinated manner which preserves the specialization of RGCs in the retina and leads to the development of a topographic map of the world represented in the OT, which is used to orient behaviors towards spatial targets. RGC innervation of the OT also has important impacts of OT development, which is the broad focus of the present thesis.

lakritz Mutation

A zebrafish pigmentation mutant was discovered in 1996 that had overexpression of pigment melanophores causing a very dark colored fish. It was dubbed *lakritz* (*lak*) which is the German word for licorice (Kelsh et al., 1996). The cause of this pigment overexpression wasn't initially known, but it was later found to be caused by a mutation in a gene causing loss of RGCs and thus blindness (Neuhauss et al., 1999, Kay et al., 2001). Zebrafish change their pigmentation based on the ambient light of their surroundings, so fish without optic nerves cannot detect the light and therefore do not retract their pigment granules.

It was further found that the *lak* mutation alters a basic helix-loop-helix transcription factor atonal-homolog 5 (also known as *atoh7*), which is an essential gene for differentiation of RGCs during development. The mutation leads to a loss-of-function allele where RGCs never differentiate in the retina, and instead, there is an overproduction of amacrine cells, müler glia, and bipolar cells. This occurs due to a delay in exit from the cell cycle when neurogenesis fails, leading to an increased pool of retinal progenitors that are able to freely differentiate into other cells during later stages of development (Kay et al., 2001).

Development and Organization of the Optic Tectum

The initial anterior/posterior patterning of the neural plate results in three distinct populations of cells corresponding to the forebrain, midbrain, and

hindbrain during the segmentation period. These structures get further divided and separated throughout the stages of development. At the end of the segmentation period, the midbrain divides to form the ventral midbrain (tegmentum) and the dorsal midbrain (optic tectum) (Kimmel et al., 1995). At this stage of development, the brain undergoes primary neurogenesis where there is a large amount of neuronal differentiation, axon tracts are established, and basic brain scaffold is set up that can produce reflexive motor responses to touch stimuli (Boulanger-Weill & Sumbre, 2019). Secondary neurogenesis occurs in the OT after 2 dpf, when RGC axons first begin to terminate in the OT neuropil.

RGC axons first cross the optic chiasm at ~32 hpf and then proceed contralaterally to their targets. A small amount of axons project to the preoptic area, but the majority project into the neuropil region of the OT around 48 hpf. After RGC axons enter the superficial tectum, they defasciculate into separate layers and then proceed to their specific targets in the OT along these layers. There are six main layers, or 'laminae' of the OT. The most peripheral lamina is the stratum marginale which doesn't receive RGC axons, then there are four retinorecipient laminae, which make up the neuropil area: the stratum opticum (SO), the stratum fibrosum et griseum superficiale (SFGS), the stratum griseum centrale (SGC), and the stratum album centrale (SAC). Lastly the most central OT layer containing most of the neuronal cell bodies is known as the stratum periventriculare, or periventricular zone (PVZ). The areas that receive the majority of retinal input are the SO and the SFGS, which is further divided into three sub laminae (Xiao et al., 2005). The initial separation of RGC axon projections into these laminae has been shown to remain specific throughout early development, where axon position between laminae does not undergo refinement. Additionally, the specific layers RGC axons project to are cell type specific, where distinct RGC combinations correspond to specific sub laminae (Robles et al., 2013).

Development and Refinement of the Retinotopic Map

RGC axons project onto targets in the OT in a spatially conserved manner, meaning that RGCs that are next to one another in the retina will connect to adjacent neurons in the OT. And RGCs on one side of the retina will project to the corresponding side of the OT. Nasal RGCs map onto the posterior/caudal OT, and temporal RGCs map onto the anterior/rostral OT (Figure 4). This results in a continuous topographic map in the OT where a complete image is able to be transferred to the brain from the eye, and then used to inform directed behaviors such as swimming, hunting or predator escape. Continuous maps are distinct from discrete maps because discrete maps involve the mapping of discrete qualities from the input field into spatial groups in the downstream fields. Examples of this type of mapping include olfaction, and taste, which involve receptor neurons corresponding to the same chemical receptors mapping onto the same discrete locations in the brain. Auditory and visual senses are both continuous maps, but vision includes elements of both continuous and discrete mapping, where RGCs form a two-dimensional continuous map in the OT neuropil but can project more peripherally and centrally depending on the type of information they encode such as ocular dominance or motion detection. (Luo & Flanagan, 2007).

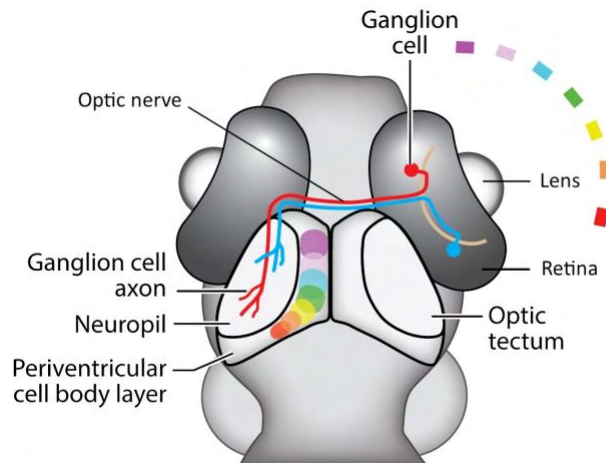


Figure 4: Topographic mapping from RGCs in the retina to the neuropil of the Optic Tectum.

Temporal-nasal and dorsal-ventral axes of the retina map to the rostral-caudal and medial-lateral axes of the tectum as illustrated, with the red RGC projecting from a nasal location to a posterior target. The rainbow colored, overlapping, circles represent how stimuli, represented by the colored boxes, map from the retina onto clusters of overlapping tectal neurons. Figure reprinted from (Bollman, 2019).

As RGC axons grow, they are guided by a combination of signaling molecules in order to reach the location of their correct targets. These molecular guidance cues are expressed by the RGCs themselves based on their position in the retina, as well as the cells they come in contact with in the OT. The most essential and well-studied guidance factors in retinotopic map development are the ephrin-A ligand and EphA receptor (Luo & Flanagan, 2007, Oster & Sretavan, 2003). These two proteins have complicated interactions with each other that inform a variety of biological mechanisms, including axon guidance. They are both membrane-bound, meaning that they require cell-cell contact to function, opposed to secreted signaling factors.

In RGC axon growth, the general mechanism involves relative levels of expression of ephrin-A interacting with relative levels of EphA expression to

either lead to axon growth cone survival to be inhibited or supported. High concentrations of Ephrin-A generally inhibit axon growth when the axon growth cone has high expression of EphA (Luo & Flanagan, 2007, Oster & Sretavan, 2003). In the visual system of zebrafish and other vertebrates with primarily monocular vision, ephrin-A is expressed highly in the posterior OT and then decreases in a gradient to the anterior OT. In the retina EphA is conversely expressed in a gradient in RGCs with nasal RGCs expressing more EphA than temporal RGCs. This sets up a counter-gradient paradigm where the high expression of ephrin-A in the posterior OT repels the axons of the highly EphA expressing nasal RGCs, leading them to terminate in the anterior OT, and axons of lowly EphA expressing temporal RGCs are less repelled by the high ephrin-A, and are able to terminate in the posterior OT. A similar counter-gradient system is set up along the dorsal/ventral axis with ephrin-B and EphB (Luo & Flanagan, 2007).

These signaling interactions are complicated slightly by both the fact that ephrins and Ephs are bifunctional, meaning that axon attraction or adhesion can occur at low concentrations, and bidirectional, meaning that Eph expressing cells can signal to ephrin expressing cells (Marquardt et al., 2005). The two protein types can also be co-expressed within the same cell or area, which is important because most regions of the CNS both receive and send projections (Luo & Flanagan, 2007).

In addition to ephrin/Eph, there are other factors involved in the formation of the retinotopic map. Wnt3 (Wingless-related integration site 3) is part of the Wnt family of proteins, mainly studied in terms of cell fate. Wnt3 has been shown to be expressed in a dorsal > ventral gradient in the OT, with a repulsive counter-gradient of Ryk receptors expressed ventral > dorsal in RGC axons, and the presence of Frizzled receptors in RGC axons caused attraction (Luo & Flanagan, 2007). Additionally, orderly and continuous mapping is achieved by axon competition. After RGC axons defasciculate in the OT, they form relatively evenly spaced arbors that fill up the available space and terminate evenly throughout the OT neuropil (Kutsarova et al., 2016).

Retinotopic Map Refinement

Initial retinotopic mapping establishes the specific OT laminae and relative position on the dorsal/ventral, and anterior/posterior axes that the RGC axons terminate in, however, termination zones are fairly broad, and this map must be refined by repositioning, pruning, and strengthening RGC/OT arbors and synapses. The major mediator of retinotopic refinement is neural activity, even though the initial map is formed properly irregardless of neural activity. Both spontaneous and evoked visual signals cause interactions between neurons and glia that lead to major structural changes in the OT (Kutsarova et al., 2016).

Long term potentiation is the process of synaptic strengthening that occurs when a presynaptic neuron and a postsynaptic neuron fire simultaneously. In the zebrafish visual system, when RGCs fire and activate OT neurons, their connection is strengthened. Additionally, OT neurons that are adjacent to each other are more likely to be activated simultaneously, and thus strengthen their connection to each other, neurons that fire asynchronously will have long term depression. The end result is an integrated network for coordinated OT responses to RGC signals. Neural activity has more effects than direct potentiation and depression, as it can also influence the signaling factors that are expressed by pre and postsynaptic neurons as well as by glia. (Luo & Flanagan, 2007, Kutsarova et al., 2016).

These mechanisms of neuronal dynamics are largely conserved between vertebrates, but there are many differences between how neurons interact and grow between groups of animals.

Adult Neurogenesis

In fish and amphibians, the visual system continues to grow past the early development period and continues throughout adulthood. This is possible due to the maintenance of stem cell populations in the retina and the OT that proliferate and differentiate concordantly to achieve coordinated growth between the two areas. This growth, along with growth from cell hypertrophy and migration,

results in an increased visual acuity throughout adulthood, giving older zebrafish an advantage in hunting over younger zebrafish (Cerveny et al., 2012).

There are populations of stem cells at the outer edge of the retina in the ciliary marginal zone (CMZ) which are maintained by continued expression of transcription factors, cell cycle genes, and signaling molecules (Wan et al., 2016, Cerveny et al., 2012). The CMZ contains different populations of cells along the peripheral-central axis, where the stages of proliferation to differentiation are like early embryonic and post-embryonic development. The most peripheral cells are multipotent retinal stem cells that can become a variety of retinal cells. These cells maintain their population through a combination of symmetric and asymmetric cell divisions, where some daughter cells become retinal progenitor cells that divide more rapidly towards the more central CMZ and differentiate into adult RGCs that are integrated into existing circuits (Valdivia et al., 2016).

After the OT undergoes primary neurogenesis, secondary neurogenesis occurs after 48 hpf, when RGCs start to enter the OT neuropil. By 66 hpf, OT neurons can respond to sensory inputs from the eye, and by 5 dpf, the zebrafish has fully functional visual circuitry and can capture prey and have a variety of visual-induced behaviors (Boulanger-Weill & Sumbre, 2019). Neurogenesis in the OT, similarly to in the retina, occurs throughout adulthood in zebrafish, however, there are differences in the progenitor pools and how neurons are integrated into existing circuitry. There are two routes that can give rise to new neurons (Figure 5). The minor route involves direct proliferation of a *her5*-positive neurogenic neuroepithelial cells pool located in the posterior part of the PVZ, close to the ventricle. This route is the most upstream and is similar to embryonic neurogenesis. The major route, which gives rise to the majority of new neurons in secondary and adult neurogenesis, involves division of *her4*-positive radial glia cells. (Boulanger-Weill & Sumbre, 2019, Galant et al., 2016). Once OT neurons are born, they are pushed into the PVZ in a ‘conveyor belt’ like mechanism rather than cell migration (Boulanger-Weill et al., 2017), and they eventually mature into a combination of GABAergic (inhibitory), and glutamatergic (excitatory) neurons, with a small percentage of cholinergic (excitatory) neurons.

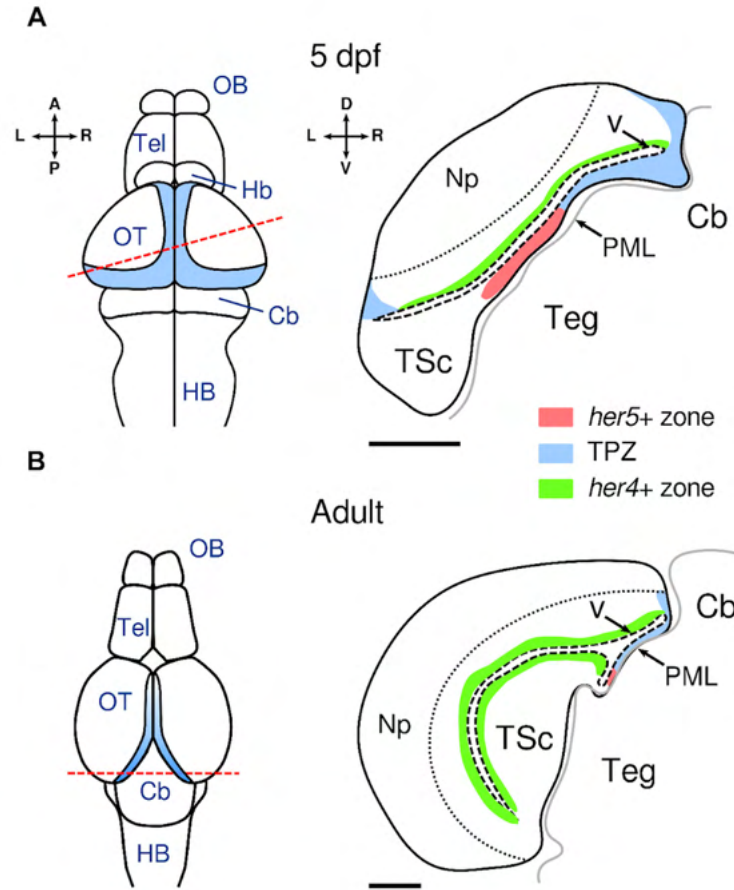


Figure 5: Proliferative zones in the zebrafish visual system

Diagram showing several different zones of proliferation in the optic tectum between the larval (A: 5 dpf) and adult zebrafish (B). The *her5+* zone represents neurogenic neuroepithelial populations. The *her4+* zone represents neurogenic radial glia cells. The TPZ represents the area where progenitors are generated that can either directly differentiate (minor route), or first generate radial glia cells (major route). Abbreviations: L: left, R: right, A: anterior, P: posterior, D: dorsal, V: ventral, OB: olfactory bulb, Tel: telencephalon, Hb: habenula, Cb: cerebellum, HB: hindbrain, Np: neuropil, TSc: torus semicircularis, Teg: tegmentum, V: ventricle, PML: peripheral midbrain layer, TPZ: tectal proliferation zone. Images from (Boulanger-Weill & Sumbre, 2019).

Oligodendrocytes

Oligodendrocytes are another glia cell that are essential for brain function in vertebrates. Their main role in adult brains is myelination of axons. Action potentials in neurons involve rapid membrane depolarization that propagates throughout the cell and its axon process. Myelin sheaths are multilayer membranes that enclose axons and decrease the membrane capacitance and decrease membrane resistance in the nodes between sheaths, which leads to fast saltatory conduction, and more efficient propagation of action potentials. Myelin also serves to protect axons from damage and maintain the axon's homeostasis and nutritional support. Diseases that affect myelination can have very adverse effects for brain functioning such as with the autoimmune disease multiple sclerosis. Oligodendrocytes have further important roles that are relatively new to being studied.

Development

Mature, myelinating oligodendrocytes differentiate from oligodendrocyte precursor cells (OPCs), which in turn, are specified from neuroepithelial progenitors (Figure 7). OPCs have different morphology than oligodendrocytes, having very arborized, space filling projections versus targeted projections that myelinate axons. There are several transcription factors known to control differentiation from OPCs to oligodendrocytes. These include Sox10, Olig1, and Myrf (Santos-Ledo et al., 2023).

OPCs

In all vertebrates, there are a variety of OPCs that never differentiate into mature myelinating oligodendrocytes. There is a lack of research currently into the functions they have in the developing and adult brain. However, we do know

that OPCs are the only known glia cell to form direct synapses with neurons (Figure 6) (Xiao & Czopka, 2023). This indicates an important function for cell-cell communication. We also know they are involved in refinement of neural circuits, and have been shown to regulate circuit refinement by engulfing synapses in developing mice (Auguste et al., 2022).

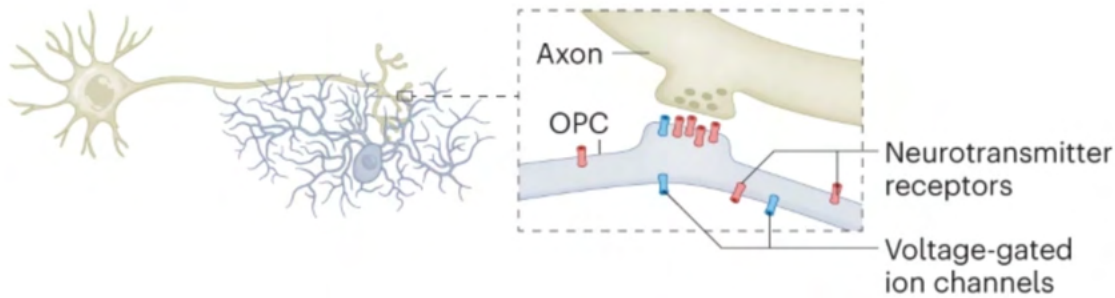


Figure 6: OPCs form synaptic contacts with neurons.

Diagram showing the synaptic relationship between an OPC and a neuron. The bottom cell is an OPC, and has highly arborized regional processes that make contact with the axons of neurons. OPCs can receive neurotransmitter signals through their receptors, that cause depolarization thanks to the presence of voltage-gated ion channels. Image reprinted from (Xiao & Czopka, 2023).

In zebrafish, there is a population of OPCs in the OT that don't differentiate into mature oligodendrocytes until after 14 dpf, and even then, the majority of them stay as OPCs into adulthood (Xiao et al., 2021). These OPCs were found to be important for the initial targeting of RGC axons into the OT. Elimination of OPCs caused the sizes of the axon arbor to be dysregulated. Additionally, these OT OPCs were found to be essential for refinement of the retinotopic map. Without the presence of OPCs during map refinement, RGC axon arbors failed to be pruned to the same level required in WT larvae. These impact of OPC ablation were found to have a negative effect on the visual acuity of the larve. However, is still unknown the mechanisms that OPCs utilize to communicate and control the mapping and refinement of RGC axons in the OT. Additionally, it is unknown how OPCs receive signals from RGC axons, and how

the relationship is coordinated (Xiao et al., 2021). The present study seeks to visualize the dynamics between the optic nerve and OPCs, and observe OPC dynamics when the optic nerve is missing.

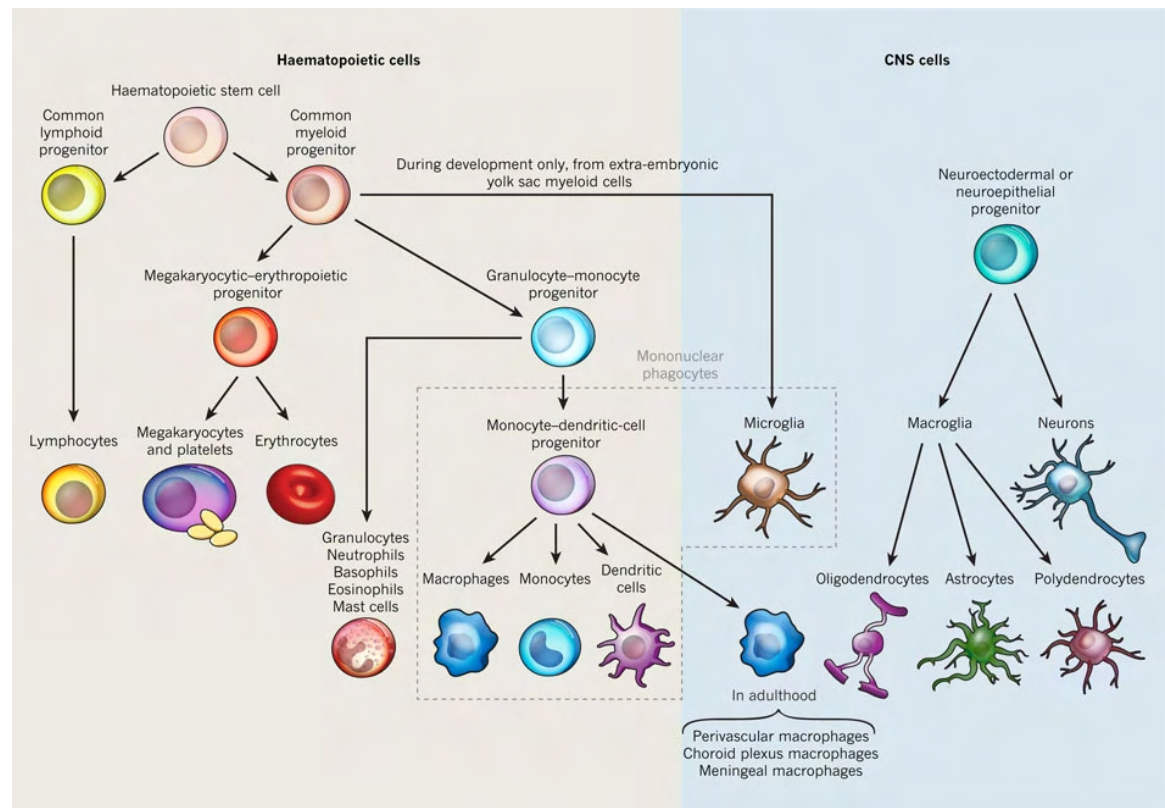


Figure 7: Cell lineages of hematopoietic and CNS glia precursors.

Microglia are derived from haematopoietic cells during development and migrate to the CNS. Other resident glia cells in the CNS are derived from neuroepithelial progenitors in the CNS. Diagram reprinted from (Ransohoff & Cardona, 2010)

Microglia

Unlike oligodendrocytes/OPCs, astrocytes, radial glia cells, and polydendrocytes, which differentiate inside the CNS from neuroepithelial cells, microglia migrate into the CNS as primitive macrophages and then become

specified as resident immune cells in the brain. They have a variety of known roles, chiefly as phagocytic immune cells. This means that they engulf and digest material such as pathogens, damage tissue, apoptotic cells, and degenerating processes. They are involved in responding to damage to the brain by migrating to the location of damage and phagocytosing the debris of apoptotic or damaged cells. (Var & Byrd-Jacobs, 2019). Similar to OPCs, they are also involved in remodeling neuronal connections by pruning synapses, and regulating synaptic plasticity and function (Hong et al., 2016).

In zebrafish, they migrate to the brain before 48 hpf (Neely & Lyons, 2021), and can be seen patrolling the OT SPV zone. They have a very ramified morphology and are very transient, traversing the whole SPV and occasionally entering into sections of the neuropil. Previous research in the Cervený lab found that microglia are phagocytic in the OT between 5-9 dpf (Jiahe Yue, 2022).

Previous research I did in summer 2023 involved studying the dynamics of microglia in relation to cells in the OT undergoing apoptosis. (Appendices A and B). The results were very preliminary, and mainly involved troubleshooting and experimenting with methodology. However, we were able to image dynamic interactions between microglia and apoptotic cells labeled with Annexin A5:GFP, which is an injected plasmid that causes cells to secrete a fluorophore that binds to phosphatidylserine on the exposed outer plasma membrane leaflet that occurs when cells undergo apoptosis (van Ham et al., 2010).

Light-Sheet Fluorescence Microscopy

Fluorescence microscopy, generally, refers to the use of fluorophores to label certain structures to be imaged under a fluorescence microscope. Fluorophores are special molecules that, when they absorb light of a certain wavelength, and then re-emit it at a longer wavelength. Green fluorescent protein (GFP) was the first fluorescent molecule that was able to be harnessed and made to be non-endogenously expressed in an organism through transgenics. It was

first purified from the *A. victoria* jellyfish in the 1960s. Its discovery heralded the development of countless variations of fluorescent proteins that can be used in genetic reporter lines, immunohistochemistry antibody stains, fluorescent dyes, etc.

In all types of fluorescence microscopy, including light-sheet microscopy, a laser of a certain wavelength is used to excite certain fluorophores which then emit light in a longer wavelength. In conventional fluorescence microscopy, the same objective is used for illumination and detection, relying on specialized dichroic mirrors that pass light of different wavelengths in opposite directions. In light sheet microscopy there are separate illumination and detection objectives (e.g., Figure 8A). In both systems, high sensitivity detectors (cameras) are mounted to capture the emitted wavelengths. In Reed's light sheet microscope (Figure 8B), there are two lasers that form gaussian beams, with a narrow waist that is the most focal point of the laser. These lasers are then moved up and down very fast, coordinated with the speed of line reading of the CMOS (Complementary metal–oxide–semiconductor) image sensor camera. This results in an effective sheet of light, hence the name 'light-sheet'. This light-sheet illuminates a very thin plane of a sample, and then the sample can be scanned through this sheet (Z plane movement), which step-by-step illuminates a 2-dimensional XY plane of the sample. These individual 2D images are then composited together to form a 3-dimensional image of a sample. Light-sheet microscopy, when configured for live cell imaging, can take these 3D stacks repeatedly over an interval in order to get a 3D timelapse of dynamics in a living organism (Wan et al., 2019).

A SiMView/MuVi-SPIM Two opposing views

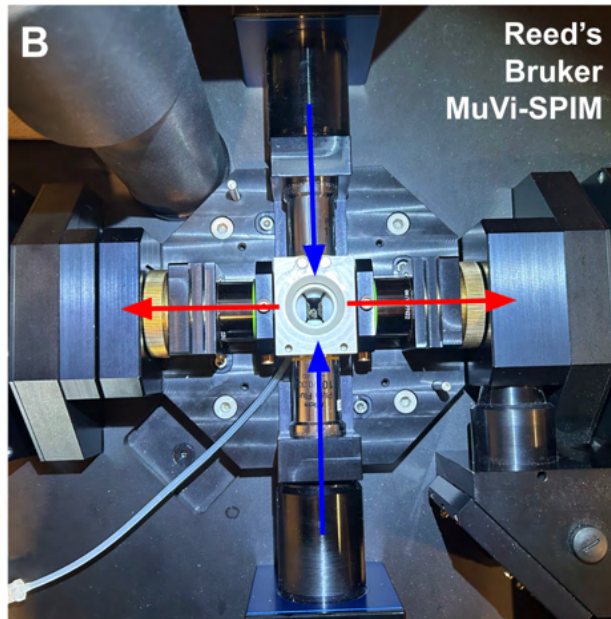
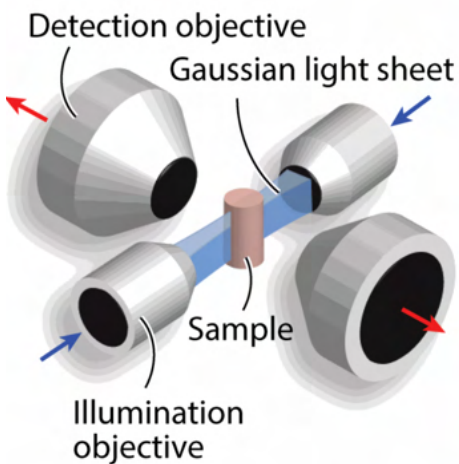


Figure 8: Dual-illumination/dual objective light-sheet microscopy

Diagram multiple-view selective plane illumination microscopy (MuVi-SPIM).

Demonstrates lightsheet microscope excitation and detection paths in two-

detection-objective, two-illumination-objective microscopes. A) shows a

simplified diagram reprinted from Wan et al., 2019. B) shows a photograph of

Reed's MuVi-SPIM from Bruker/Luxendo.

Present Study

How does retinal innervation influence glial cell dynamics in the zebrafish optic tectum? Specifically, how does a lack of the optic nerve effect the dynamics, movement, and localization of microglia and OPCs?

There is a large derth of research on OPC functions, and there are very few studies characterizing their morphology and behaviors in zebrafish. My first set of goals in this regard were to see, in WT larvae from 2-9 dpf, where OPCs are localized, how their processes are organized, how they interact with RGC axons and with eachother, how they develop and migrate within and around the OT,

and any other interesting or useful observations on their morphology and dynamics.

In the context of past research in the Cervený lab that was focused on retinal innervation dependent development of the OT, my second goal was to see how all of these aspects of OPCs from 2-9 dpf might be different when there is no retinal innervation (*lak* mutation). There wasn't much past research with which to base a hypothesis on, so both sets of goals were largely exploratory.

In addition to studying OPCs, I was interested in how microglia act without the presence of the optic nerve. We know from past research that microglia are phagocytic in the zebrafish optic tectum from 5-9 dpf, and that they contain a large number of lysosomes (Appendix B; Figure 19). However, we don't know which cells they are phagocytosing, or have a comprehensive understanding of their repertoire of behaviors and functional interactions with other cell types. I had the additional goal of visualizing microglia behavioral dynamics, and the differences between WT and *lak* mutants between 5-9 dpf.

Methods

Transgenic lines

Tg(atox7:RFP) – poggi et al. 2005: Labels retinal ganglion cells (RGCs), including their projections that make up the optic nerve.

Tg(nkx2.2a:GFP) – Kucenas et al. 2008: Labels a variety of cells with GFP including oligodendrocyte precursor cells (OPCs). *Nkx2.2a* is also known to be expressed in a variety of neural progenitor cells, including neural progenitors in the ventral forebrain and midbrain and spinal cord as well as a subset of pancreatic cells (Kucenas et al. 2008, Pauls et al., 2007).

Tg(mpeg:GFP) – Ellett et al., 2011: Labels macrophage lineage cells with GFP, including body macrophages, and microglia in the brain.

Tg(mpeg:mCherry) – Ellett et al. 2010: Also labels macrophage lineage cells, including body macrophages, and microglia in the brain but with the fluorescent protein mCherry instead of GFP.

Zebrafish Husbandry

Adult transgenic fish lines are maintained on a 14-hour light/10-hour dark cycle at 28°C. They are fed twice daily with a mix of dry fish flake food, 1-day hatched brine shrimp and rotifers depending on their age. Transgenic embryos were obtained by pairing male and female fish from respective lines in breeding boxes separated by a removable divider. Around 9 AM the next day, the divider would be removed to allow the fish to breed at a specific time.

Rearing Embryo/Larvae

One day after collecting the embryos/injecting them, they were checked, and all unfertilized or dead ones were discarded and the E3 was changed. When the embryos were 3-4 dpf, they would either hatch on their own, or I would remove the chorion and change the E3. Between 3-5 dpf, the larvae could be screened for the level of transgenic expression using a fluorescent dissecting scope. Annexin V:GFP expression could be screened easily because the plasmid included cardiac myosin light chain 2 also driving GFP (*cmlc2:GFP*), a fluorescent myocardium that made the heart glow bright green, so if they had high transgenic expression, then the hearts were bright. *Atoh7:RFP* could be screened by looking for fluorescent eyes, optic nerve, and optic neuropil. Homozygous *lak* mutant larvae could be screened by looking for *Atoh7:RFP* positive larvae that didn't have optic nerves. Fish carrying *Tg[nkx2.2a:GFP]* could also be screened easily because successfully transgenic animals had bright green fluorescence all throughout the nervous system due to the abundance of oligodendrocytes primarily in forming white matter. The *mpeg:GFP* marker could be screened by looking for individual green fluorescent spots throughout the larva that represent individual microglia cells. However, the *mpeg:mCherry* marker was more difficult to screen for due to the dimness of the mCherry protein, potentially also coupled with difficulties due to my colorblindness.

Larvae were fed rotifers starting at 5 dpf, and then raised to either five, seven, or nine dpf and then imaged live using light-sheet microscopy.

Imaging

File Naming Conventions

Throughout the steps of imaging, processing, and analysis, I stuck to certain naming and organization conventions for my files to include the necessary

information in as little space as possible. This was done by forming an acronym that corresponds to a specific larva at a specific time point. The acronym is structured as follows:

First is the experimental group (WT is **W** / *lakritz* is **L** / enucleated is **E**), then type of experiment or main transgenic marker (Oligodendrocyte is **O**, microglia is **M**), then additional transgenic markers (annexin A5 is **A**, *atoh7* is **At**), then I would put the number corresponding to which experiment I had done (the first experiment I did in any experiment type is **1**, the second experiment is **2**, etc.), and then I used a descending letter of the alphabet to indicate the individual larvae in order (the first larva I imaged was **a**, the second was **b**, third was **c**, etc.), and then I put a number corresponding to the days post fertilization of the larva at the time I imaged it (five dpf was **5**, seven dpf was **7**, etc.).

Examples: **WOAt2c7** means the wildtype larva expressing the oligodendrocyte marker and *atoh7* marker from my second experiment, and it was the third individual larvae I imaged from the experiment (c corresponds to 3), and I imaged it when it was 7 days old. **LMA1a5** was the *lakritz* larva expressing the microglia and *atoh7* markers in my first experiment, and it was the first individual larva I imaged from the experiment, and I imaged it when it was 5 days old. I imaged some larva over multiple days, meaning the images I took at one dpf might be **WOAt3b2**, and the next might be **WOAt3b3**, but they are both included in the **WOAt3b2** file path because individual images/timelapses are contained in their own folders within another folder, and it is simpler, and better for understanding the contiguity, to have one main folder for a given larvae and set of images/timelapses.

Prep

At five, seven, or nine dpf, individual larvae were selected to be imaged. Selection for ideal imaging was based on the sufficient expression of transgenic markers, as well as a relative lack of pigment and observed health. I placed the selected larva into a separate small petri dish and anesthetized it with 1x tricaine

(0.0015% w/v). While still in the main lab room, I prepped the agarose to embed the larva by placing a tube with a ~0.3 mL aliquot of agarose and microwaving until melted. I also prepared a 15 mL tube filled with 1x tricaine and 1x PTU in E3 to use as the fluid in the imaging chamber. I then transported the agarose tube, chamber fluid, and larvae dish to the room housing the light-sheet microscope, where the chamber fluid could be poured into the imaging chamber, and the agarose tube could be placed into a 55°C heat block. After checking that all the microscope, computer, and accessory power strips were turned on, and the acquisition software was opened, the larva could be mounted.

Mounting

To prep the capillary tube for mounting, the tubes first had to be cut to length with the diamond glass cutter and the dedicated trimming tool. I then inserted the blue plunger wire and cut it such that about a centimeter of wire was sticking out of the rougher cut edge of the capillary tube. Then, a small amount of agarose was sucked up into the capillary tube by inserting the end of the tube into the agarose and pulling back on the plunger. This agarose was then allowed to flow along the sides of the wire and dry slightly. This was done in order to form a better plunger seal, and so that the agarose doesn't flow through the tube after the larvae is mounted.

Once the capillary tube was prepped, I placed the larva on a dry petri dish lid and removed most of the water from around the larva. Then I added a large drop of agarose on top of the larva, after being cooled slightly in the pipette. Then, before the agarose could solidify, I used the capillary tube to suck up the larva. This was achieved by first drawing up several millimeters of agarose into the tube, and then pushing the tube onto the tail of the larva and pulling up the rest. The end position of the larva in the tube was important for the quality of the image later on, so I would typically reposition the larva slightly by pushing back out part or all of it and changing the angle that it was drawn up from until it was at a sufficiently parallel angle to the tube. Once the larva was in place, the tube

was placed horizontally and allowed to cool, making sure that the agarose at the tip of the capillary was level with the tip (not forming a meniscus), this was to ensure that agarose at the tip didn't dry too fast and stick to the tip, which would then squeeze or reposition the larva when it was pushed out.

After the agarose was cooled enough to keep its shape when pushed out of the tube, it was pushed out until the entire head of the larva was sticking out. Then the end of the plunger-wire sticking out the back of the capillary was trimmed off with wire cutters. The larva was then finished being mounted and ready to be placed into the imaging chamber.

Laser Calibration

I typically calibrated the lasers while the agarose was cooling. If the microscope had just been powered on, some of the calibration parameters were typically quite off base and needed more alignment. If there is an error message indicating that the photomanipulation interlock is open, then there is a button on the back of the machine near the power plug that needs to be pressed to override the error. I adjusted the stage all the way to the lowest x value to get it out of the path of the lasers and made sure I was in the calibration view in the software. Then, I parked the laser, uncoupled the offset, set the beam expander to 1.6 μm , and turned the 488 nm laser on and set it to 100% power. I set the filter wheel to the 580-627 nm filter, because it somehow resulted in a much clearer and stable laser than the 497-554 filter and then hit the "start live" button in the image viewer window. The live view should then show two boxes with a single beam going horizontally through the center.

To align the lasers, I first blocked one of the sides by sliding a business card between the beam tube and the octagon. I adjusted the uncovered laser offset until it was centered on horizontal crosshair, and then adjusted the y value until the waist of the beam was around 1.5 tick marks from the middle vertical line, and then the z value until the beam was as in-focus as possible. I then switched the business card to block the other laser path and repeated the

alignment for the other laser. Once both lasers were focused and aligned, I stored the current offsets and rechecked the box to couple them, and then started with the “line calibration” box, making sure to let the viewer window update completely before hitting “next” through the steps. After calibration, the mode should automatically change to line mode.

I made sure to check the larvae in the tube every couple minutes, and if the agarose was solidified enough before I finished calibrating the lasers, then I simply placed the tip tube into the fluid filled petri dish. I raised the stage all the way back up, lowered the laser power to ~10% and inserted the capillary tube larva side up in the stage holder, clicked the “start live” button again, and lowered it back down until the head of the larva was visible in the frame.

Image Acquisition

Once the larva’s head was visible in the frame, I used the stage controller to rotate the larva until both eyes/both tectums were on the same Z plane, and the dorsal part of the midbrain was closest towards the camera I was using. I then adjusted the Y and X values until the anterior tip of the midbrain was close to the bottom edge of the screen, and the larva was centered. I then set up the stacks by moving dorsally along the Z axis until the OT was just out of view and clicking “new stack”, and then moving ventrally along the Z axis until I was just past where I wanted to image. I then sometimes set up additional rotation stacks by repeating a similar process after rotating the stage. I then cropped the acquisition image by clicking the box in the upper right of the image and dragging in until the posterior part of the larva, and the extra space around the sides was cropped. Cropping out unwanted parts of the image during acquisition can be very important for reducing file sizes. I continued to set up the channels by raising the number of lines in the camera box from 20 to ~60, setting the exposure time to 75-100 ms, setting the laser power to ~6-15%, making sure the laser that’s on matches the filter wheel, and then clicking “add” in the channel box.

Then, I added an “event”, “task”, and “trigger” to the boxes on the right and filled in the stacks and channels I wanted to image and the timelapse intervals and number of repeats depending on the experiment. For OPC experiments, I did 10-minute intervals for an extended period of time (up to 24 hours). For microglia experiments, I did 2-minute intervals also for extended periods of time, though the fluorescence in the microglia, especially when labeled with mcherry, is dimmer and tends to photo-bleach more than the OPCs and thus can’t effectively be imaged as long.

I changed the sample tracking parameters so that the largest shift is 50 μm , and the smallest shift is 5 μm , and started the tracking. Then, after double checking that all of the channels and stacks were correct, I hit “run” to take the image/start the timelapse.

Tradeoffs

In the image acquisition step, there are some tradeoffs that can be adjusted depending on the desired outcome:

There is a tradeoff between laser power and exposure time versus photobleaching. Increased laser power and exposure time can increase the brightness and clarity of the image, but it causes photobleaching and potentially slight toxicity.

The beam splitter can be adjusted to change the laser itself. The 1.6 μm beam splitter has narrower waist but it more spreads throughout the edges of the sample. The 3.0 μm beam splitter has a more even axial distribution, but it doesn’t converge into a bright waist, so the overall image is generally less intense. I imagine that the higher beam splitters work better in clearer samples when you want a more even axial illumination.

Changing the number of lines

Choosing the time interval between images should be based on the expected dynamics of the cells you’re looking at, I wanted to be able to track cells that are faster moving like microglia with images taken every two minutes, while

OPCs migrate much slower, so a 10-minute interval was about sufficient, but in a future experiment looking at the growth of processes in sparsely labeled OPCs, a shorter interval would likely be needed

Rotations increase isotropic resolution, but double/triple file size, and processing with rotations decreases overall resolution a lot, but they can be excluded during processing or processed separately to have high XY resolution images of different views. In general, rotations aren't worth it for later time points that look good in dorsal to ventral stacks. This tradeoff also applies to imaging using multiple cameras. When imaging a single dorsal to ventral stack, all that is needed is a single camera, and using a second camera only serves to increase the file size and confuse the processing later.

Cropping and reducing the file size is important. It might seem like a trade off, but for as long as the data you are looking for isn't compromised, it is a generally good practice. A smaller and less complicated file is much easier to work with in image processing and analysis. Large files are computationally taxing, which increases processing and loading times and can crash the computer. Cropping during acquisition is best, but you can also crop the images by setting a region of interest (ROI) in the Luxendo image viewer during processing, or by cropping the 3D image in Imaris, which is sometimes necessary when there is extra blank space around the sample that is added during processing.

Image Processing

The Luxendo image processing software contains many parameters that can be freely adjusted and switched on and off according to the type of data you're working with or how you want the image to be. I am still unsure on a lot of what the specific parameters are doing, and a lot of my time was spent troubleshooting and experimenting with different processing parameters, but there are certain steps that are more consistent or straightforward with the

process. To process an image or a timelapse, I often viewed the data through other means first to determine how I should process it.

Maximum Intensity Timelapse

I would usually use a simple python software script coded by Jan Frankowski, our Bruker representative. This software converted each acquisition file to .tiff output representing a maximum intensity projection of the image stack (a single 2D image, where every pixel corresponds to the brightest pixel from all of the Z planes). This results in a series of .tiffs in a folder that can be dragged and dropped into ImageJ/Fiji and played as a fast timelapse. This step is much faster than processing the whole 3D image, and allows you to watch the timelapse at a fast speed without needing to wait for Imaris, the more comprehensive 3D analysis software, to load the processed images. This lets you make sure that the larvae stayed in view the whole time and allows you to determine the relative health of the larvae throughout the course of the timelapse. If I noticed that the larva died throughout the timelapse, or its tissue swelled or declined in clarity, then I knew to crop the video during processing to a timepoint before there were any signs of unhealthiness. Because they are played at a high speed, the maximum intensity projection timelapses can also be used to observe overall cell patterns in the timelapses that occur on a large timescale (like over the course of the whole movie vs between a few frames).

In addition to temporal cropping, if the initial acquisition stack wasn't as cropped as I wanted it, or if I had a specific goal of looking at a very small window (like in Figure __) then I might need to set up a region of interest (ROI) in the image viewer window before setting up the actual image processing.

Luxendo Image Processor

In terms of actual image processing steps, there was a lot of variety in the parameters I experimented with, but the images I have included in this thesis were fairly consistent. Its also important to note that many of the parameters are

very unintuitively described, and difficult to grasp what they actually do. Many of them can be selected and deselected at will without having an effect on the image unless the acquired image has specific aspects to it that require extra processing.

In the Luxendo image processor, the first step was to create a new processing task, set the “Usage level” to 2 and then click the “Browse” button next to “Imaging file paths”. Then I navigated to the folder that was created when I hit run in the image acquisition step. This folder should contain the “raw” folder which has all of the raw image files from every stack/channel combination for every timepoint. Then I changed the name of the processing task in the box next to “Task description”. I usually named it by adding the unique larva acronym (Ex. LOAt2b7); then adding the length of the timelapse I would be working with if it was a movie, or if it was a single image, I would add “preimage” or “poster” instead; then adding a note about the processing resolution I would be using (Ex. hiXYres for high resolution in the XY plane but lower isographic Z resolution, or midres, etc.). So the final title would be something like “LOAt2b7-12.6hr-hiISOres”. I then navigated down and deselected any rotations or channels that were acquired during acquisition but weren’t necessary for the processing goals, and I changed the length of the movie as needed by adjusting the end time point.

I then went to the next “Pipeline Steps” tab “Registration”. Many of the main registration boxes could actually be deselected for my purposes. Many of my images involved using a single camera to capture a single stack with no rotations, sometimes even using just one channel, which means nothing needed to register. But if using two cameras, then the main “Objective, camera registration” box needs to be checked, along with the “Register also different objectives” box, and the “Maximum translation” box can be deselected. Additionally, if using two different rotations, or stacks that need to be stitched together, or a stack that has multiple tiles, then the “Tile registration” box should be selected, and the “Re-register each time point” box should be selected, especially if the sample moved at all during the imaging process. Additionally, “Views are pre-aligned” should probably be deselected, because the stage-rotation calibration is often skewed. Many of the parameters in the “Acquisition” tab seem to not have an effect on the image if they are selected but are actually

unnecessary, however, deselecting unnecessary parameters seems to speed up the processing speed significantly.

The next Pipeline Step is “Fusion”. The “Fuse different cams” box should be checked if using both cameras. The “blending” box is involved in overlapping the images from the two cameras or from different stacks/rotations with each other to create a more continuous stitched image. I generally found more success with image resolution with this box unchecked. This also goes for the “Image-quality weighting” box, which is involved in weighting chunks of the images from different cameras and rotations to prioritize a higher quality view of a given region over a lower quality view of that same region (Ex. prioritizing the clear view from a closer camera over the blurry dim view of the further camera). However, having these parameters selected usually resulted in a choppy image that doesn’t look good from any angle. These downsides in the fusion tab are generally why I found it to be better to process different rotations separately if I wanted to get a clear view of the cell structures from multiple angles. I never really succeeded in processing an image that had ideal isotropic resolution and looked good in multiple planes of viewing. Additionally, the more elements (channels, cameras, rotations) included during acquisition, the larger the file size, and the more of these that are included during processing, the more computationally taxing the final image is in Imaris. I hypothesize, however, that later image analysis in Imaris using the algorithms might be more effective when there is a better isotropic resolution.

The last pipeline step tab in the image processor, “Output” is the most straightforward. The sliders in the Down/up-sample factors box are where you can adjust the resolution. The “Width, Height” value is simply the resolution of the XY plane. A value of 1-1.2 was generally sufficient for ideal clarity of details, and a value of 1.3-1.6 was used more if the speed of loading and watching through a timelapse was more important than getting the best image. A value of greater than 2 can still be fine up to a certain point to make out the cells and their substructures. The “Depth (non-isotropic)” checkbox, if left unchecked, will transfer the same resolution of the XY to the Z (AKA it will be isotropic). If checked, then you can adjust the Z resolution. I typically set this value to 1.5, or

sometimes higher if I was just doing qualitative analysis by scanning through the Z planes using a slicer. If I was experimenting with the isotropy or doing quantitative analysis, then I would either deselect the box, or set it to a smaller value like 1.3.

All of the other default parameters in the pipeline were left untouched. After completing these steps, checking that the parameters were how I wanted them, and checking that I had renamed the task description, I hit the run task button and waited for it to complete the processing. Then I opened the processed images in Imaris and checked that the processing worked well.

Image Analysis

Analysis of the OPC timelapses was done mainly qualitatively. After fine tuning the processing steps to optimize the tradeoff between resolution and computer loading speed/ability, I would load the timelapses and familiarize myself with the geography of the labeling. I would also adjust the display channel to get the maximum amount of clarity for both channels. I found that a helpful way to adjust the channels was to turn up the maximum until you can make out the details within the brightest parts of the image and it's not just overexposed, and then turn up the gain also until you can make out the details in the dimmer parts of that you're interested in, and then turning up the minimum to between 50-150 or until most of the background fluorescence is gone, but you can still make out the dim details of interest. After making sure the details and adjustments were as clear as feasibly possible, I would watch the timelapses over and over at different z depths using the ortho-slicer or oblique-slicer tools in Imaris set to different thicknesses. This allowed me to observe the migration and movements of cells, the growth of cell processes, interactions between cells, and cell dynamics such as mitosis, apoptosis, and phagocytosis.

Manual Spot Creation

In order to quantify the number of OPC cell bodies in the OT (Figure __), I used the spot surface creation tools in Imaris. I manually went through each Z-plane of the image and used manual spot creation to put a spot at every visually identified OPC cell body. They were easy to identify visually despite the dense labeling, but the Imaris algorithms could not distinguish a cell body from the surrounding processes. I then simply inputted the number of spots in each OT lobe in each larva into an excel spreadsheet (drive.google.com – Cervený lab “Optic Tectum” folder – “Kaelin 2023-2024 Summer & Thesis” folder – “Kaelin – Thesis Quantitative Analysis” google sheets file).

Surface Creation

To quantify the volume that the OPCs occupy, I went through the Imaris surface creation wizard. The first step was to click the blue surface button to make a new surface, and then deselecting “Segment only a Region of Interest”, then clicking to the next step. I used absolute intensity thresholding with 1 μm smooth surface detail, which is relatively large, because I didn’t need to capture tons of details of the OPC processes. Then clicked to the next page and I gradually moved down the threshold minimum value (left number), until all of the space in the neuropil that the OPC cell bodies and processes took up was covered by the surface. I also selected the box to enable split touching objects. Without this, any section of surface that is touching another section will be treated as the same surface, but with it enabled, the algorithm uses “seed points” that are added in a certain threshold based on the relative intensity in a region in order to make a lot of surfaces that are separated from one another. This is necessary in many cases, and was necessary in this case because the OPC processes in the ventral OT neuropil connect seamlessly into the rest of the large ventral body of *nkx2.2a*⁺ cells, so there couldn’t be one continuous surface that covered all the OPC volume in the neuropil that didn’t connect into a surface also

covering the rest of the fluorescence in the image. From this step, I clicked to the next page set the seed point minimum threshold to a value that included quite enough seeds that were adjacent to each other in regions that I wanted separated (between lateral/dorsal neuropil and skin, and between the ventral neuropil and rest of ventral structures). I then clicked to the next page which creates the surfaces and lets you filter them through various parameters. In this case, I didn't need to filter surfaces because I would be manually deleting all of the ones not in the OT. I then clicked to the next page which lets you assign surfaces to different classes and then deleted the default filter because I didn't need any classes, and then I clicked the double right arrow at the bottom to finish out the creation wizard.

After the surfaces were created, I had to go through and delete all of the ones that weren't in the neuropil. To delete surfaces, I went to the "edit" tab in the surface window, and then selected a surface or many surfaces and clicked "Delete". My process involved first deleting the bulk of random surfaces way ventral or posterior to the OT, and then zooming in to select surfaces finely around the neuropil, often switching between the 3D and 2D slicer view by clicking the button in the top right of the surface window labeled, when you hover over it, "Turn on/off slicer rendering for selected object". I also paid attention to the surfaces around the skin, and made sure to delete the segments that were mostly just labeling skin versus OPCs. When everything but the neuropil OPC surfaces were deleted, selected all of the surfaces in one neuropil side, and clicked the "Unify" button in the "Edit" tab. I repeated this for the other side, and then clicked onto the "Statistics" tab, and went to the individual selections tab and wrote down the Volume number in a spreadsheet (drive.google.com – Cervený lab "Optic Tectum" folder – "Kaelin 2023-2024 Summer & Thesis" folder – "Kaelin – Thesis Quantitative Analysis" google sheets file).

Masking out a channel

If an image had a region that has bright fluorescence that obscured the area I was interested in (Ex. if the skin had autofluorescence like in *mpeg:GFP* images, or the ventral *nkx2.2a+* structures were very bright), I often masked it out in Imaris. This involved going through the surface creation steps as above, but filtering out all small surfaces except the one large surface that hopefully only covered the region I wanted to mask out. Then, once the surface was created, I navigated to the “Edit” tab in the surface window and clicked on the “Mask All” button, which opens a window. I then selected the channel that the thing I wanted to mask was in, and then unselected the box next to the “set intensity outside surface” slider, and selected the box next to “set intensity inside surface”, and made sure it was at 0. This duplicates the channel and sets a version where the whole region inside the surface is deleted.

Microglia Analysis

There were quite a few images and timelapses that included microglia labeled with either GFP or mCherry, however, I was only able to analyze one WT and one *lakritz* from the 5 dpf group. This was because I wasn’t able to get any *mpeg:GFP* images at 7 dpf, and the 9 dpf ones were all under an hour and not very high quality. Additionally, I had a few *Tg[mpeg:mCherry]* images/timelapses from research during summer 2023, but even with the light sheet microscope, the mCherry appears to photobleach very fast, and the timelapses I took during this time had a time interval of every 5 minutes, rather than every 2 minutes, so comparison within groups would have been difficult here.

In terms of the 5 dpf analysis, I processed both static images and extended 4 hour timelapses. For the static images, I first masked out the skin, which had a lot of unwanted fluorescence, using the steps detailed earlier. Then, I went

through the normal surface creation wizard, thresholding so that the details of the microglia processes were visible. I filtered out all surfaces that had fewer than 8360, and greater than 116449 voxels, because there were small surfaces that formed around some mysterious background labeled cells, and some large surfaces that formed around leftover skin background, or adjacent macrophages outside the OT. I then set up two classes of cells. Class A was made of cells with a sphericity value less than 0.695, and class B was made of cells with a sphericity value greater than 0.695. The sphericity value of 0.695 was an arbitrary cutoff where cells seem to appear more spherical than not. I then completed the surface creation and manually deleted all of the surfaces that were outside of the OT. I then went into the statistics tab and wrote down in an excel spreadsheet (drive.google.com – Cervený lab “Optic Tectum” folder – “Kaelin 2023-2024 Summer & Thesis” folder – “Kaelin – Thesis Quantitative Analysis” google sheets file) the average volume, and sphericity, as well as the overall number of microglia, and the number of microglia in each sphericity class.

The timelapse analysis was similar to the static image analysis. However, I didn't mask out the skin, and there were the added surface creation steps involving “Tracks” which are representations of the surfaces across the dimension of time. In the track sections of the surface creation, I left the parameters at default, except for the “Track Duration” filter, during the “Filter Tracks” page, which is set to include only tracks with a duration above 40 minutes, to make sure there weren't any small tracks from random surfaces that were created from background fluorescence throughout the movie. After the surface/track creation was complete, I selected all of the tracks that existed outside the OT and deleted them. Then I went to the statistics tab, and wrote down in a spreadsheet the average (per time point) volume, sphericity, overall number of microglia, and number of microglia in each sphericity class. I also wrote down the average displacement length value at the last time-point, and the track speed variation (drive.google.com – Cervený lab “Optic Tectum” folder – “Kaelin 2023-2024 Summer & Thesis” folder – “Kaelin – Thesis Quantitative Analysis” google sheets file).

Qualitative analysis of these images largely remains to be done. I'm not sure if there are potential patterns of movement or behavior that are different between groups that can't be quantified

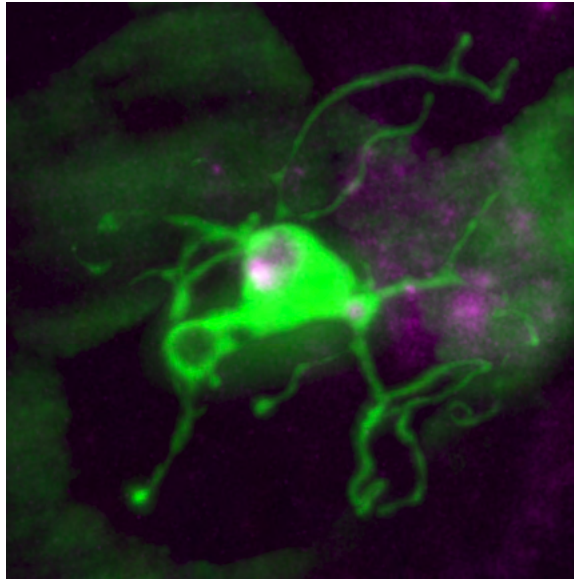


Figure 9: Demonstration of a surface macrophage with reduced tissue depth-related aberration

Imaging deep into tissue results in increased light scattering and distortions, causing optical aberration. This image shows what is likely a macrophage (non CNS microglia), that was residing on the surface of the skin in one of my images. It showcases very fine diameter processes. Its likely that much of the observed morphology of microglia, and other cells with dynamic fine processes, is effected by distortions from the depth of tissue.

In developing zebrafish larvae, the optic nerve crosses the optic chiasm around 32 hpf, and then enters into the OT around 48 hpf (Robles et al., 2013). Based on timelapses I took towards the beginning of the year, it seemed like at around this time-point, OPCs were entering into the OT directly ahead of RGC axons, but I wanted to capture some better views of early 2-3 dpf development. When imaging that early, there are added difficulties in mounting the specimen because the larva's head is curved more around the yolk sac, which prevents the light-sheet microscope's ability to capture a dorsal view of the OT. However, in

my last OPC experiment, the larva (WOAt3b2) ended up tilted to the left after mounting, and this surprisingly ended up offering a great lateral view of the left OT of the larva. This timelapse ended up becoming a large focus of my qualitative analysis for this project. Figures 10, 11 and 12, are all acquired from the image acquisition session of this specimen.

Results

Nerve Growth and Nerve Coordinated OPC migration

In the lateral view of the aforementioned high resolution timelapse of the 2-3 dpf WT larva (WOAt3b2), I was able to observe a lot of the timing and presumed cellular interactions between RGC axon growth, and OPC migration. The timelapse acquisition was started at around 5:38 pm, which would put the larva at around 55 hpf. However, the speed of zebrafish development is variable and highly temperature dependent, so it's likely this larva was immature and its developmental age was much younger than its actual age.

The video, which has been cropped to just the lateral part of the midbrain, starts out with a several large connected structures labeled by *nkx2.2a:GFP*, including the pre-optic tract and a ventral bulk of *nkx2.2a* + cells (Figure 10 A), and there are *nkx2.2a* + projections that start to leave from the around the meeting point of the preoptic tract and ventral area. These projections grow dorsally and medially in an arc. They have growth cones, and radial glia-like morphology. Then, the first pioneer RGC axons enter into this tectal frame area between timestamp 05:00 (hh:mm) and 08:50 (Figure 10 B). These sparse early projections send out highly transient exploratory growth cones. They seem to first veer into the pre-optic tract and then retract and continue along past the pre-optic tract. At this same time point, there are a few *nkx2.2a* + cells with a round morphology that become visible within the pre-optic tract, and they then migrate up the tract to occupy a dorsal/medial position.

More RGC axon projections grow and come in contact with the pre-optic tract/ventral area by timestamp 10:30. The early RGC axons all travel medially around the pre-optic tract. Then, by time 14:20, the first OPC cell bodies can be

distinguished from the rest of the *nkx2.2a*+ structures, and by time 15:20, they can be seen sending out processes towards the tectum (Figure 10 C). By time 18:50, there are about 3 OPCs traveling towards the OT. They have highly arborized multi-polar projections that are very asymmetrical towards the direction of the OT. These early OPCs are traveling alongside the early medial optic nerve fibers, the furthest edges of the OPC projections are about neck-in-neck with the furthest edgest of the RGC axons. By time 24:00 (the end of the timelapse), there are about 7 OPCs that are actively migrating towards the OT (Figure 10 D). There are two visually distinct layers of nerve fibers, with the OPCs mainly associated with the medial layer, but by this time, many of the OPC processes are close to the lateral layer as well.

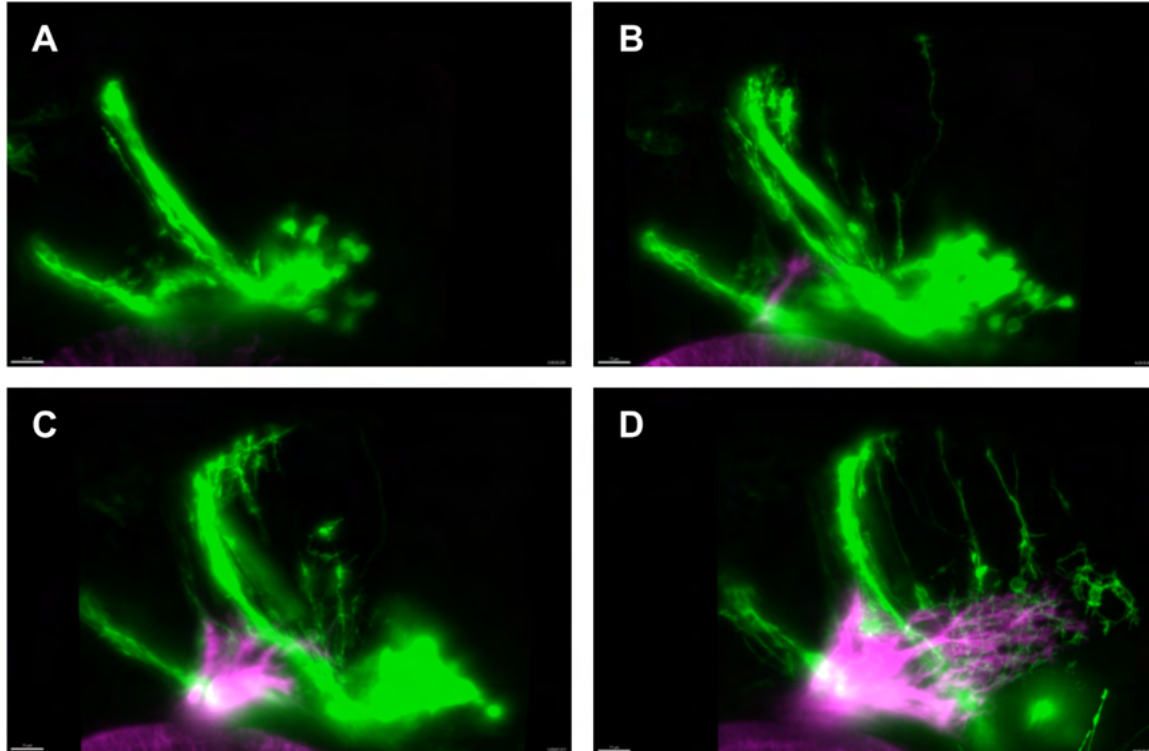


Figure 10: 2-3 dpf time-course of optic nerve growth and OPC migration

Figure A, B, C, D: Still images taken from a live-cell timeplapse starting at 2 dpf, and ending at 3 dpf in a WT larva. Lateral view of a single side of a larval zebrafish midbrain. Cells expressing *nkx2.2:GFP* fluorescent marker are labeled in green. Cells expressing *atoh7:RFP* fluorescent marker are labeled in magenta. Scale bar is 30 μm.

Optic Nerve Growth Patterns

As mentioned in the earlier section, the optic nerve splits into two distinct sections after crossing the pre-optic tract. The early RGC axons migrate medially and back around the tract, while the slightly later growing axons migrate along the lateral side of the tract (Figure 11). These layers continue to remain separated as they travel deeper into the OT. It's known that the optic nerve separates into specific laminae when it enters the tectum and continues in these laminae until they reach their targets. It's likely that the medial layer are the axons that innervate the SFGS neuropil lamina, and the lateral layer are the axons that innervate the SO neuropil lamina. These likely aren't novel findings, but it is interesting to look at optic nerve growth as it relates to other structures. Specifically, it is apparent from these images that the *nkx2.2a*⁺ projections that appear to make up the pre-optic tract might have some bearing on the defasciculation and separation of the optic nerve.

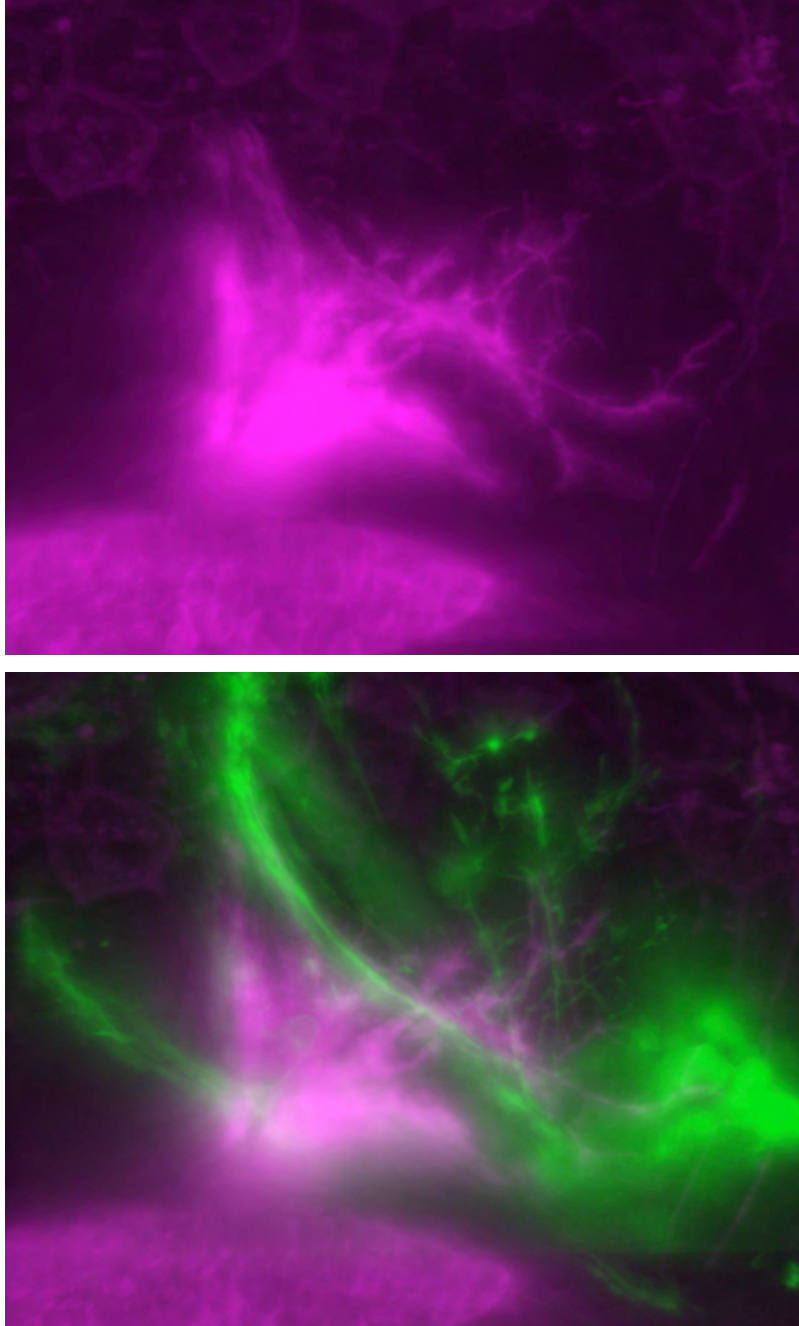


Figure 11: Defasciculation of the optic nerve in relation to the pre-optic tract

These are lateral images of the left side of the midbrain in a 2 dpf WT larva. Magenta is *Tg(ato7:RFP)*, labeling RGCs in the top and bottom images. Green is *Tg(nkx2.2a:GFP)* labeling OPCs and various structures in the midbrain in the bottom image. Both top and bottom images are of the same region at the same scale. The optic nerve encounters the pre-optic tract and early projections veer around the medial edge, with some sending projections into the pre-optic tract.

The *nkx2.2a* Reporter Labeling Specificity

The *nkx2.2a:GFP* reporter is known to label neural progenitors that differentiate into OPCs and oligodendrocytes, as well as OPCs and oligodendrocytes themselves (Kucenas et al. 2008). It has also been shown that *nkx2.2a* is expressed in a variety of cells/tissues, such as neural progenitor cells in the ventral forebrain, midbrain, and spinal cord, as well as in certain pancreatic cells (Kucenas et al. 2008, Pauls et al., 2007). However, we don't know all of the cells that express this reporter in zebrafish, and there appears to be a variety of cell types and regions labeled.

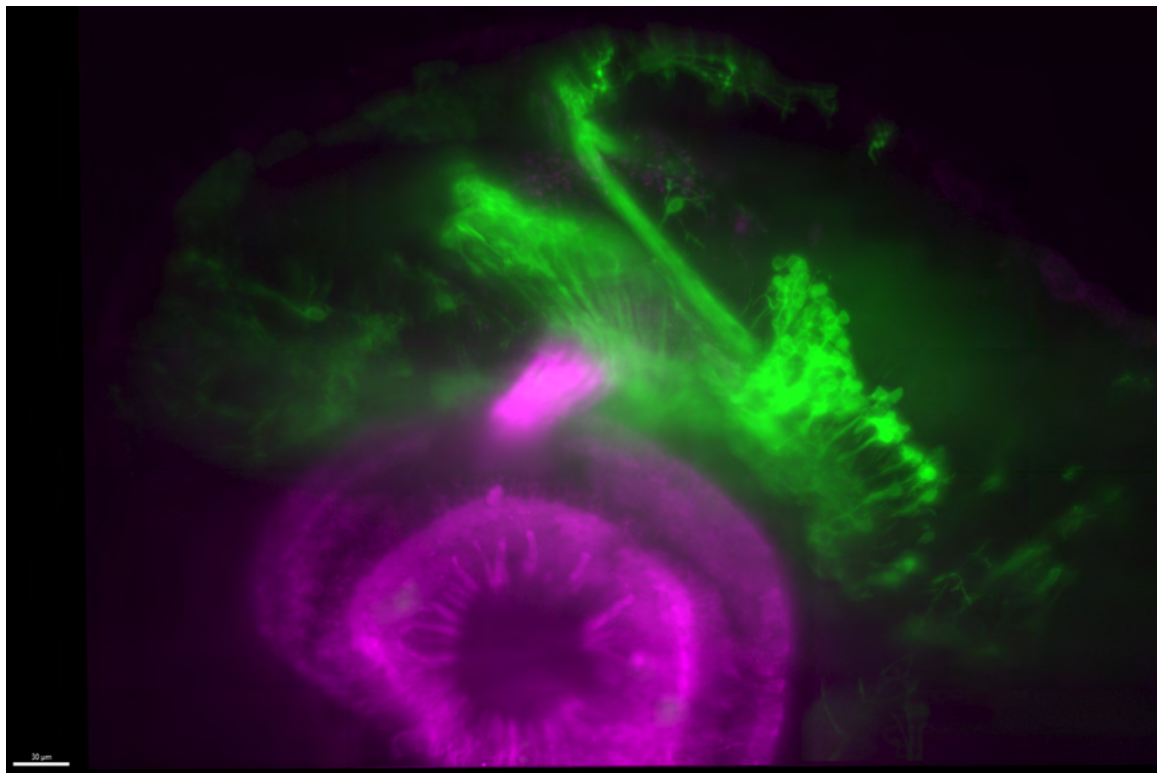
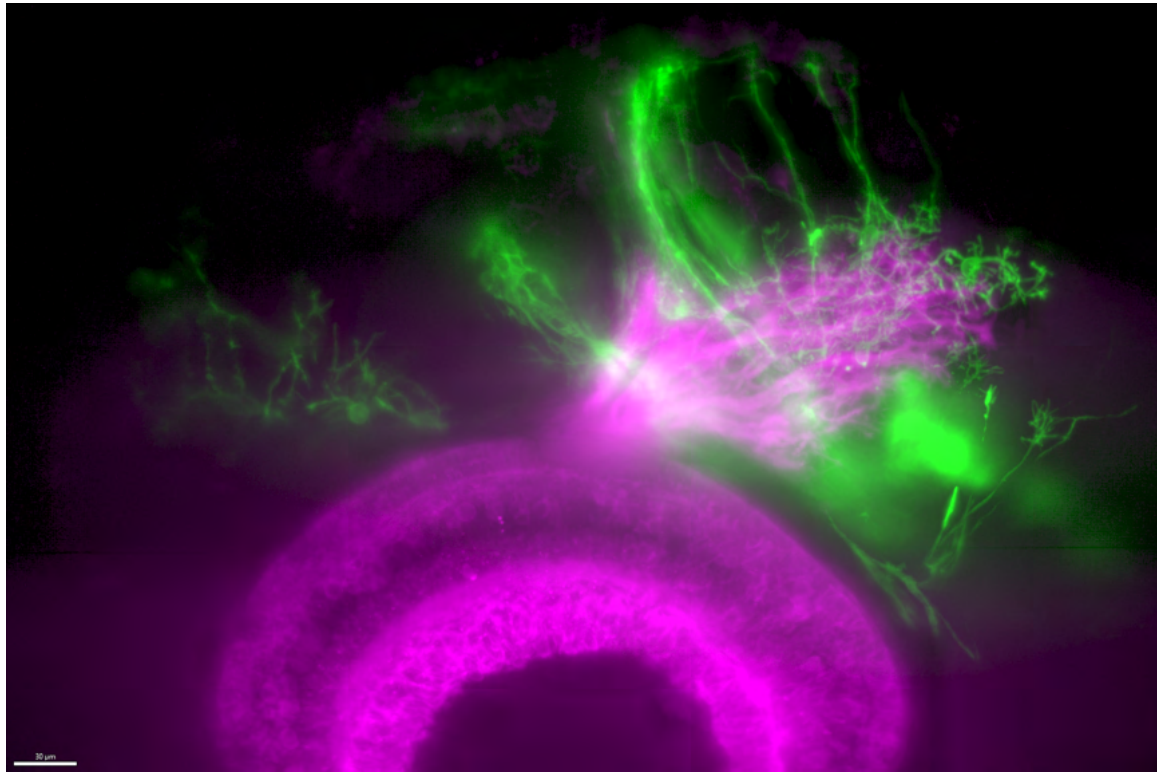
In 3 dpf larvae, I have observed seven distinct structures, cell populations, or cell types labeled by *nkx2.2a:GFP*. The first of which is the main focus of this thesis: OPCs in the OT. These can be seen migrating from the ventral area of the midbrain, adjacent to the ventricle, to the neuropil region of the OT starting at 2 dpf and continuing mainly until 5 dpf. They have the expected OPC morphology with highly arborized multi-polar processes that occupy their own space in the neuropil (Figure 12; top). I also observed a population of OPCs that migrated anteriorly, towards the forebrain. These OPCs emerged from a similar, but slightly more medial location than the OT OPCs.

Another highly dynamic cell type I observed were the long projections that bridge the two halves of the tectum. They appear early in development, likely before 2dpf, and remain very prominent at 3 dpf, and more sparsely by later 5-9 dpf. These cells have radial glia-like morphology, with very long intertectal projections (Figure 12; top). Many cell bodies are visible along the midline, with projections reaching down into both sides of the tectum. The length of these projections travels along the medial edge of the basket-shaped structure of OPCs observed in the OT. This means they likely travel around the boundary between the OT PVZ and neuropil.

In terms of midbrain structures visible with *nkx2.2a+* cells. I observed the pre-optic tract, the tegmentum (Teg), and the torus semicircularis (TSc), as well as a dense body of cells throughout the ventricular area (Figure 12; middle). The

morphology of cells in the Teg and TSc is mainly monopolar, with a single process extending down towards the ventricle.

Additionally, I was surprised to find a *nkx2.2a*⁺ cell in the retina (Figure 12; bottom). This is before any OPCs are expected to travel to the eye. This single cell strongly resembles an amacrine cell, which is a type of interneuron in the retina with a cell body residing in the inner nuclear layer, and asymmetrical processes that face into the inner plexiform layer. This description matches the lone cell perfectly.



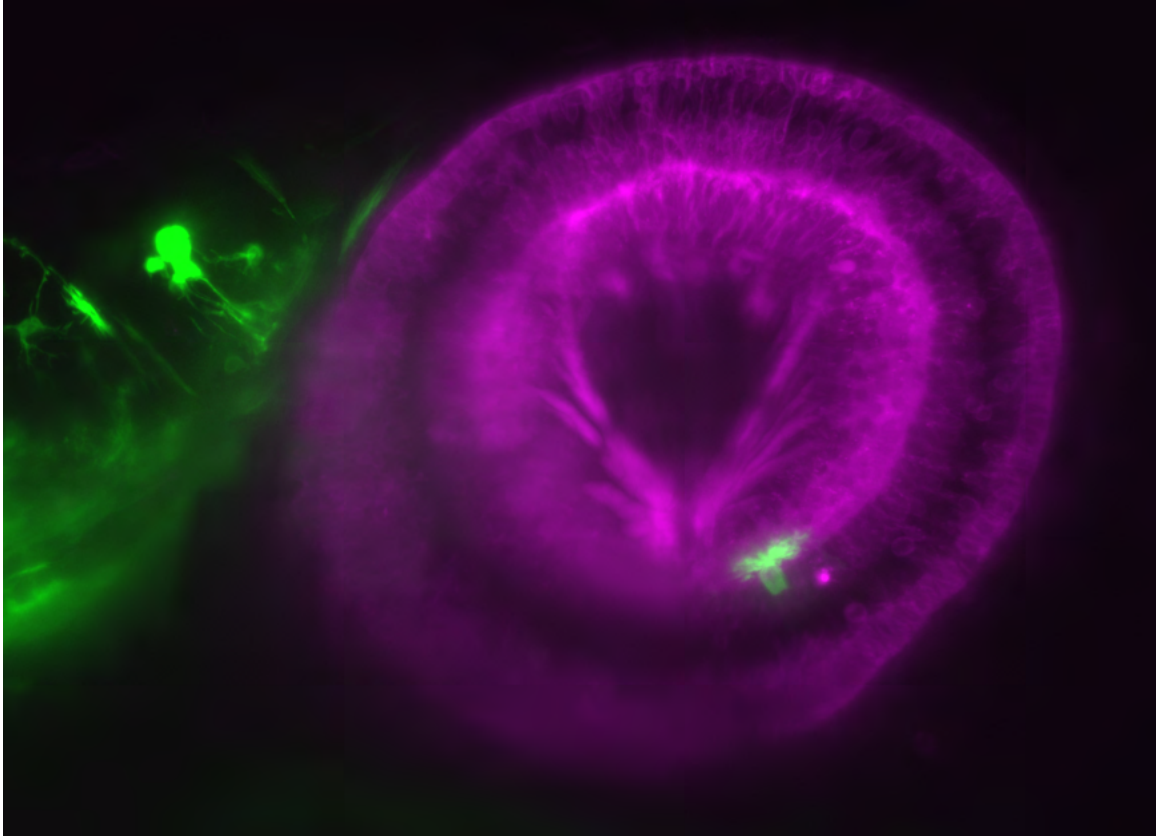


Figure 12: Differences in the range of cells and structures expressing *nkx2.2a:GFP*

All images are different lateral views of a 3 dpf WT larva. The scale bar for the top and middle images is 30 μ m. The scale for the bottom image is similar. Magenta is *Tg(ato7:RFP)*, labeling RGCs. Green is *Tg(nkx2.2a:GFP)* labeling OPCs and various CNS structures. The *top* image is a more lateral plane, and demonstrates the morphology and localization of OT OPCs (middle/right of the image), as well as forebrain OPCs (middle/left), and radial glia-like cells (top/right). The *middle* image is a more medial plane, and showcases midbrain structures with cells expressing *nkx2.2a*, including the Teg (central/left), TSc (central/right), and optic-tract (central/upper). The *lower* image is a view of mainly just the eye, and shows a single *nkx2.2a*⁺ cell in the retina that resembles an amacrine cell.

Early *lak* and WT Qualitative OPC Comparison

By 3 dpf, the general organization of the OPCs in the OT has begun to be established. There are several OPCs with their cell bodies located at the boundary between the OT PVZ and neuropil. They have asymmetric projections facing the neuropil that are already highly branched and space-filling (Figure 13; top). This structure is similar in *lak* mutants, but there are fewer OPCs, and the distinction between the neuropil and rest of the tectum is less clear because there are no RGC axon terminals to populate the neuropil (Figure 13; bottom). When observing the OPC behavior over time, OPCs appear to migrate into the main OT structures for both groups, but there appears to be more migrating cells in *lak*. Additionally, one of the migrating OPCs in a WT timelapse can be seen projecting a long process towards the OT, but then, it undergoes apoptosis blebbing and fragmenting.

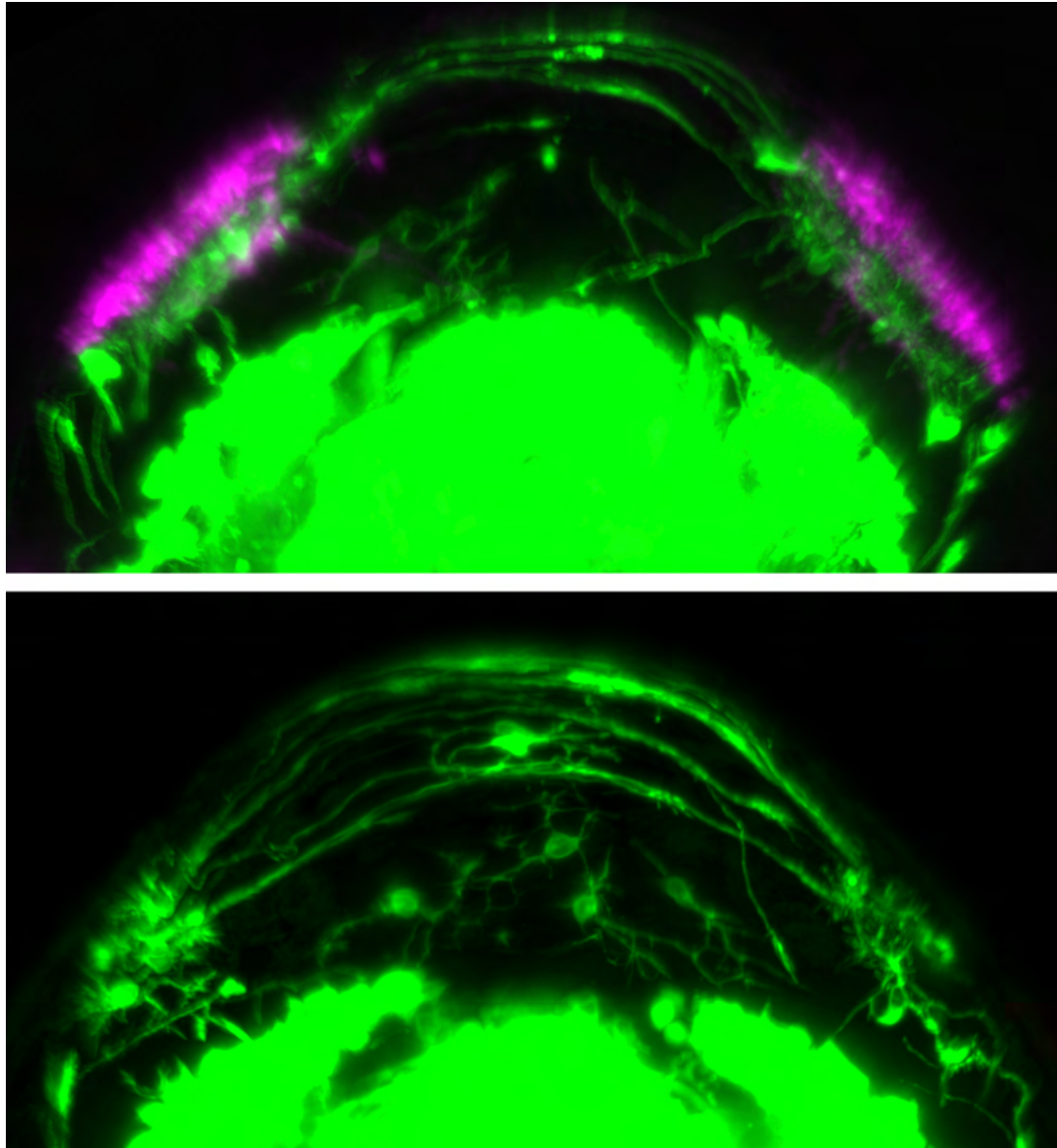


Figure 13: Comparison of WT and *lakritz* OPCs at 3 dpf

Both images are the same horizontal/dorsal views of the 3 dpf OT, at the same scale. The top image is a WT larva and the bottom image in a *lak* larva. Magenta is *Tg(ato7:RFP)*, labeling RGCs. Green is *Tg(nkx2.2a:GFP)* labeling OPCs. The bottom image has no Magenta because *lak* mutants have no RGCs.

By 5 dpf, the WT OPC structure in the OT is well established. There is a main basket-like structure with OPC cell bodies primarily located at the PVZ/neuropil boundary, with neuropil-facing projections, but there are also OPCs more associated with the RGC axons outer neuropil layers, and they form a

counter-structure with asymmetrical projections facing medially in towards the inner neuropil/PVZ. There are also many relatively sparse projections that reach into the PVZ from the basket OPCs (Figure 14; top). The *lak* mutants at 5 dpf display some variation to their OPC structure morphology. OPC projections seem to be less uniformly asymmetrical and organized, and they appear to be missing many of the cells associated with the more peripheral neuropil. The overall OPC structure also appears to be flatter and smaller (Figure 14; bottom)

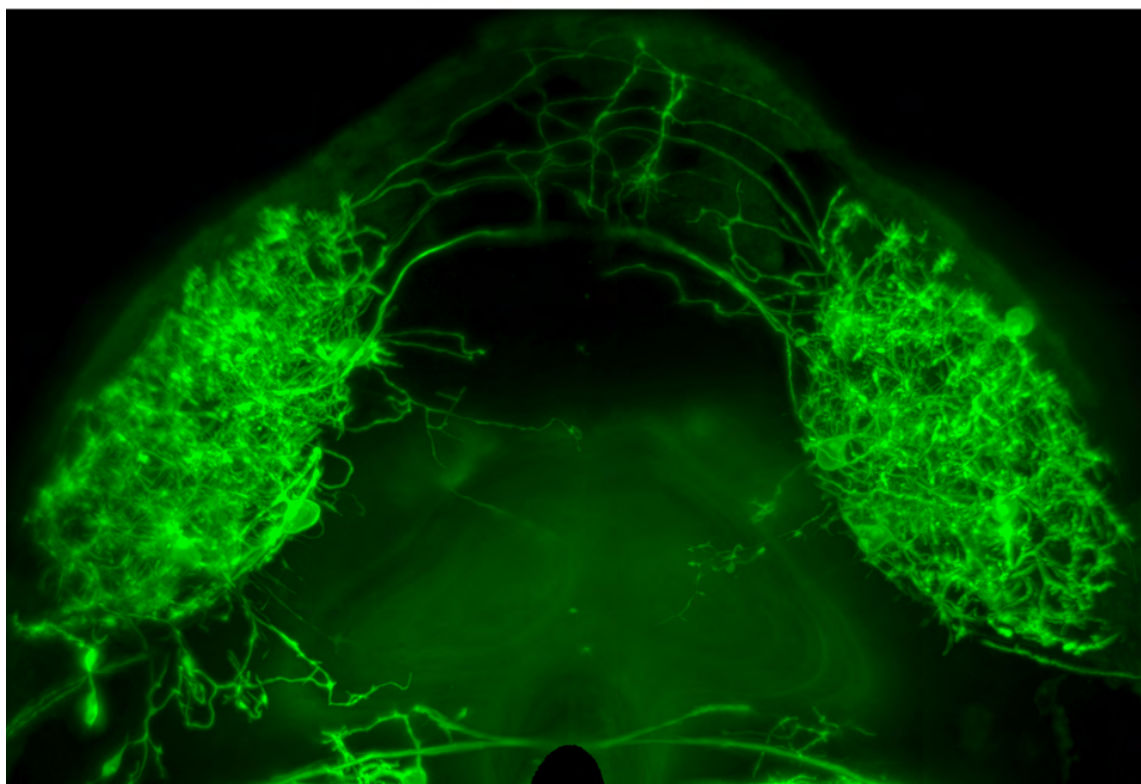
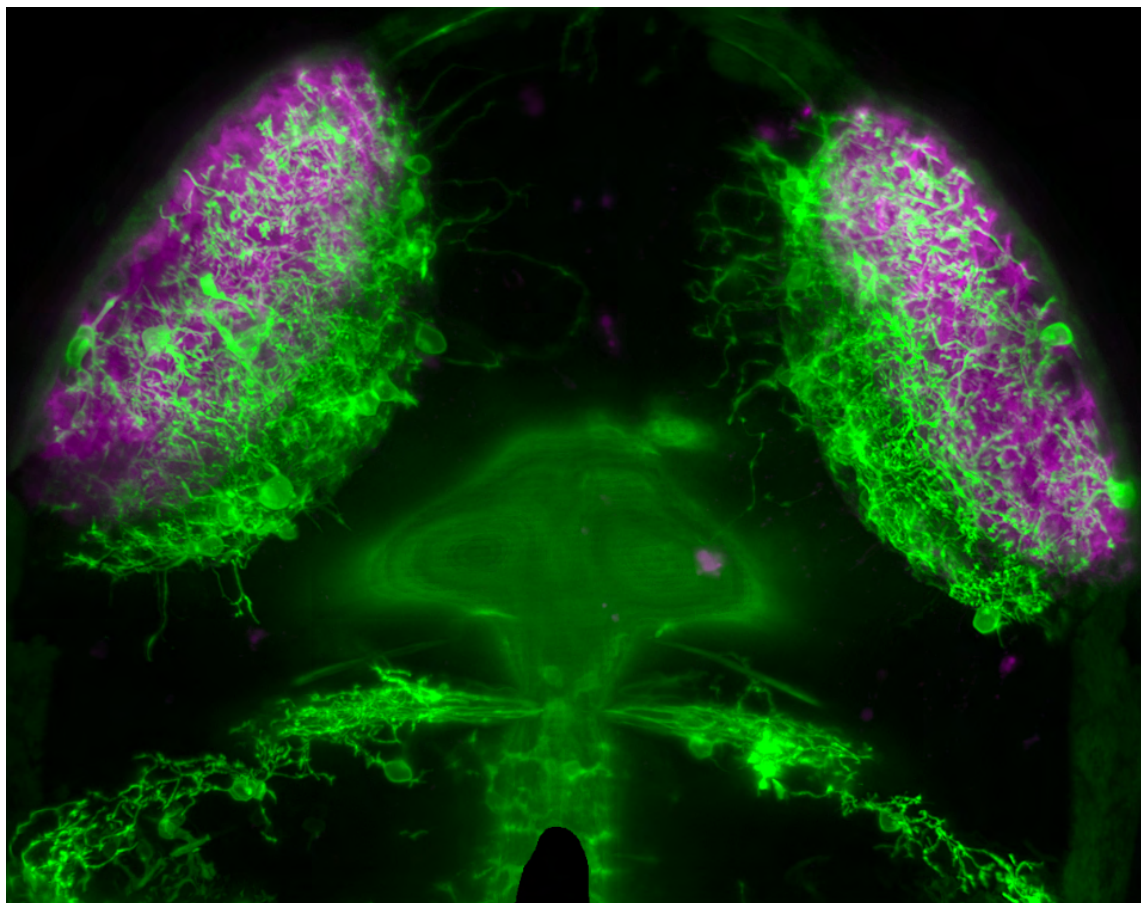


Figure 14: Comparison of WT and *lakritz* OPCs at 5 dpf

Both images are dorsal views of the 5 dpf OT, at the same scale. The top image is a WT larva and the bottom image in a *lak* larva. Magenta is Tg(*atoh7:RFP*), labeling RGCs. Green is Tg(*nkx2.2a:GFP*) labeling OPCs. The bottom image has no Magenta because *lak* mutants have no RGCs.

WT and *lak* mutants have differing OPC cell count and morphology

The qualitative observations about the morphology and cell number were able to be quantified. In terms of cell number, WT larvae appear to have a relatively stable number of OPC cell bodies from 5-9 dpf, with between a 100-120 cell bodies present in the OT for each time-point. The *lak* mutants, in comparison, have much fewer OPC cell bodies, especially at 5 dpf. However, there is an increase in cell bodies from 5 to 7 dpf, and from 7 to 9 dpf.

The volume that the OPCs occupy follows a slightly less linear pattern though, in both WT and *lak*. In WT the OPC volume at 5 and 7 dpf is similar, but is much higher at 9 dpf. In *lak*, the volume is smaller overall for every time point, but the 7dpf timepoint showed the largest volume, followed by the 9 and then 5 dpf.

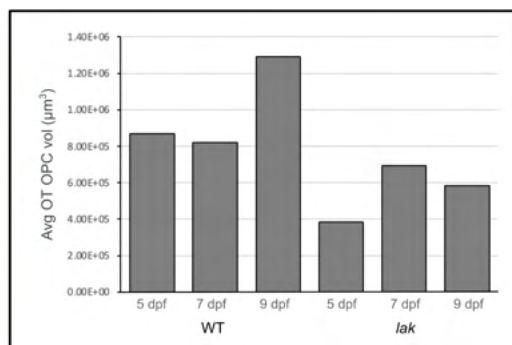
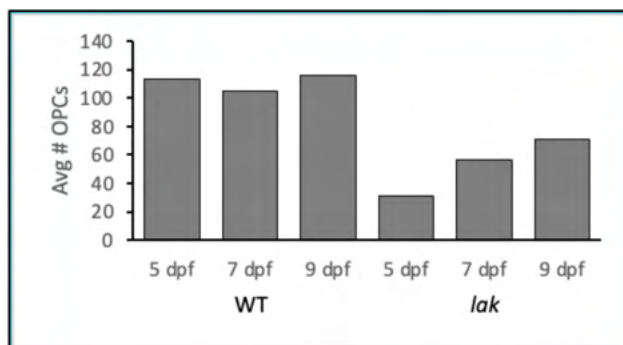
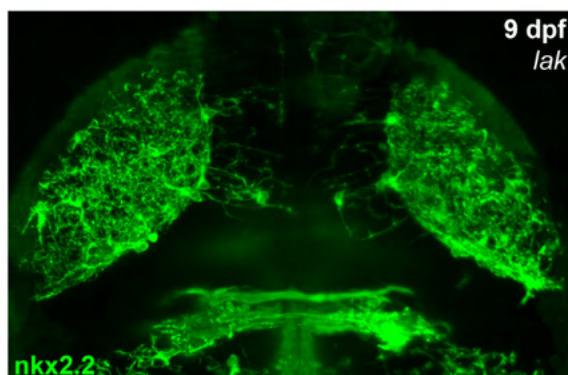
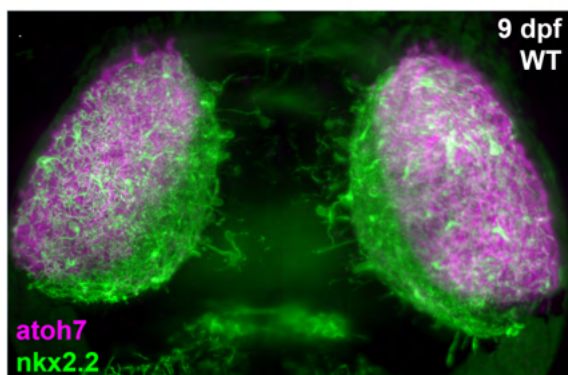
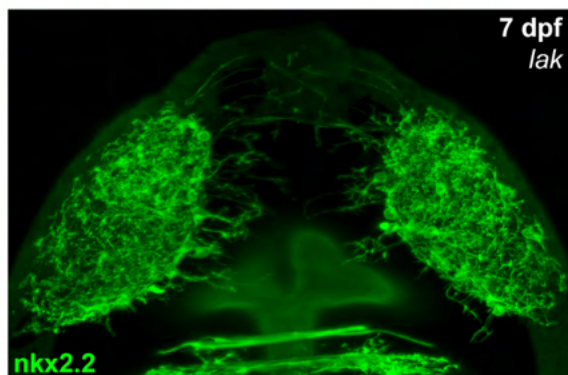
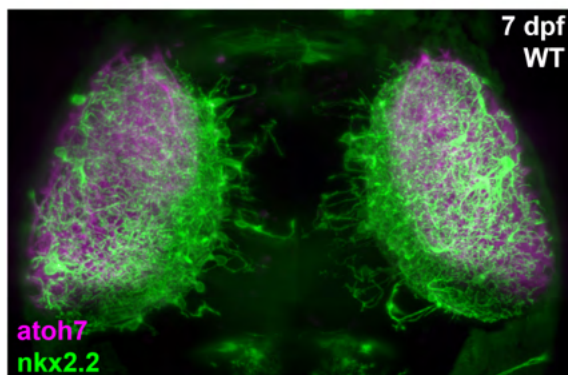
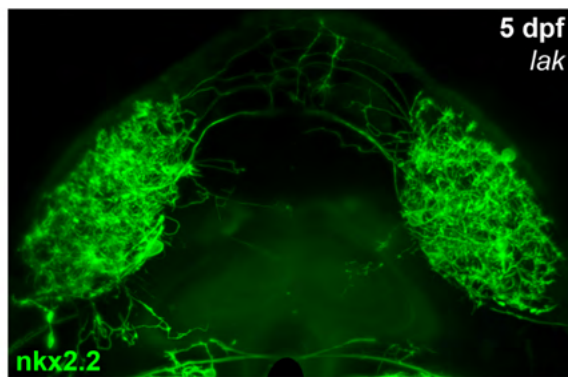
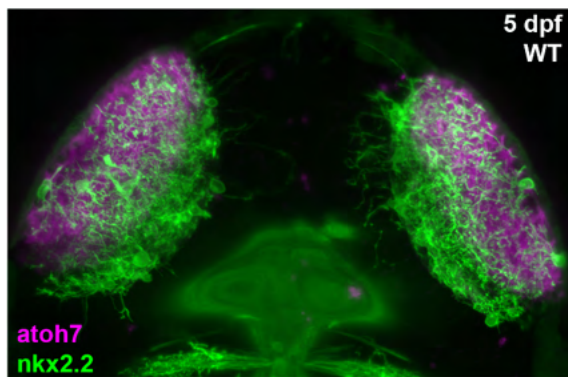


Figure 15: Quantification of WT and *lakritz* OPCs at 5, 7, and 9 dpf

All images are dorsal views of the larval zebrafish optic tectum. WT larvae images are on the left, and *lak* larvae images are on the right. All images are roughly the same scale. Magenta is Tg(*atoh7:RFP*), labeling RGCs. Green is Tg(*nkx2.2a:GFP*) labeling OPCs. The images on the right have no magenta because *lak* mutants have no RGCs. Bar charts at the bottom depict the average number of OPCs (left) and the average volume occupied by tectal *nkx2.2+* cells (right) for the ages and genotypes indicated on the x-axes.

Lightsheet live timelapses show a variety of dynamic microglia behaviors

Microglia were observed undergoing apoptosis, phagocytosis, and mitosis at 11 dpf in *lak* larvae. Apoptosis occurred in one previously healthy looking microglia (Figure 16A), and then immediately after it started blebbing, another microglia sent out several processes to make contact with the first microglia (Figure 16B), and then retracted them right before sending out a single process that engulfed the apoptosis microglia (Figure 16C). Then the second microglia moved the rest of its cell body to fully engulf and phagocytose the first microglia (Figure 16D). In another larvae, a relatively static microglia can be seen sending out multi-polar projections in all directions (Figure 16E), before retracting them and forming a spherical ball (Figure 16F). It then can be seen undergoing telophase and dividing into two symmetrical cells (Figure 16G), and it results in two microglia with normal morphology that immediately start patrolling the area (Figure 16H).

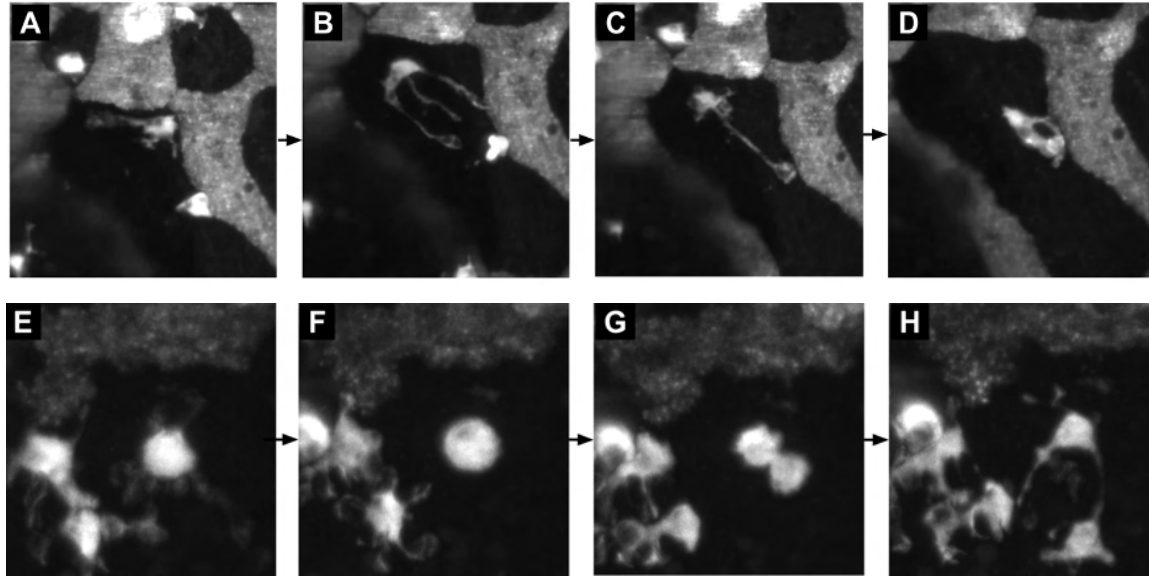


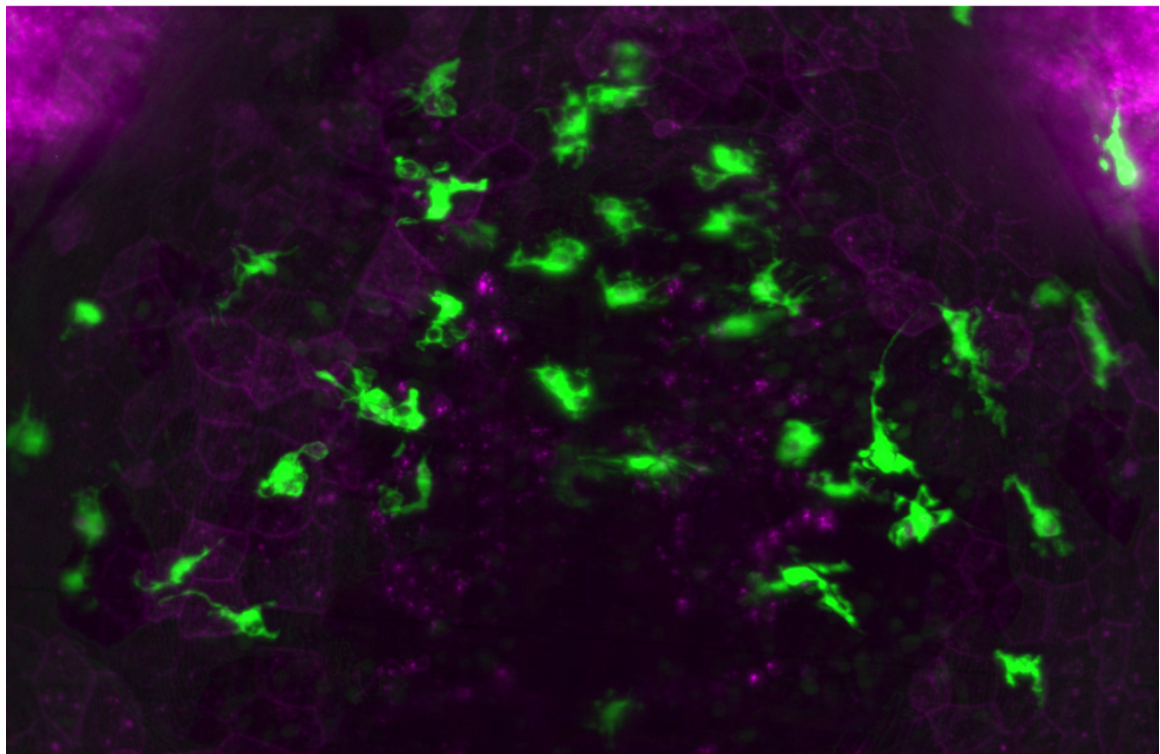
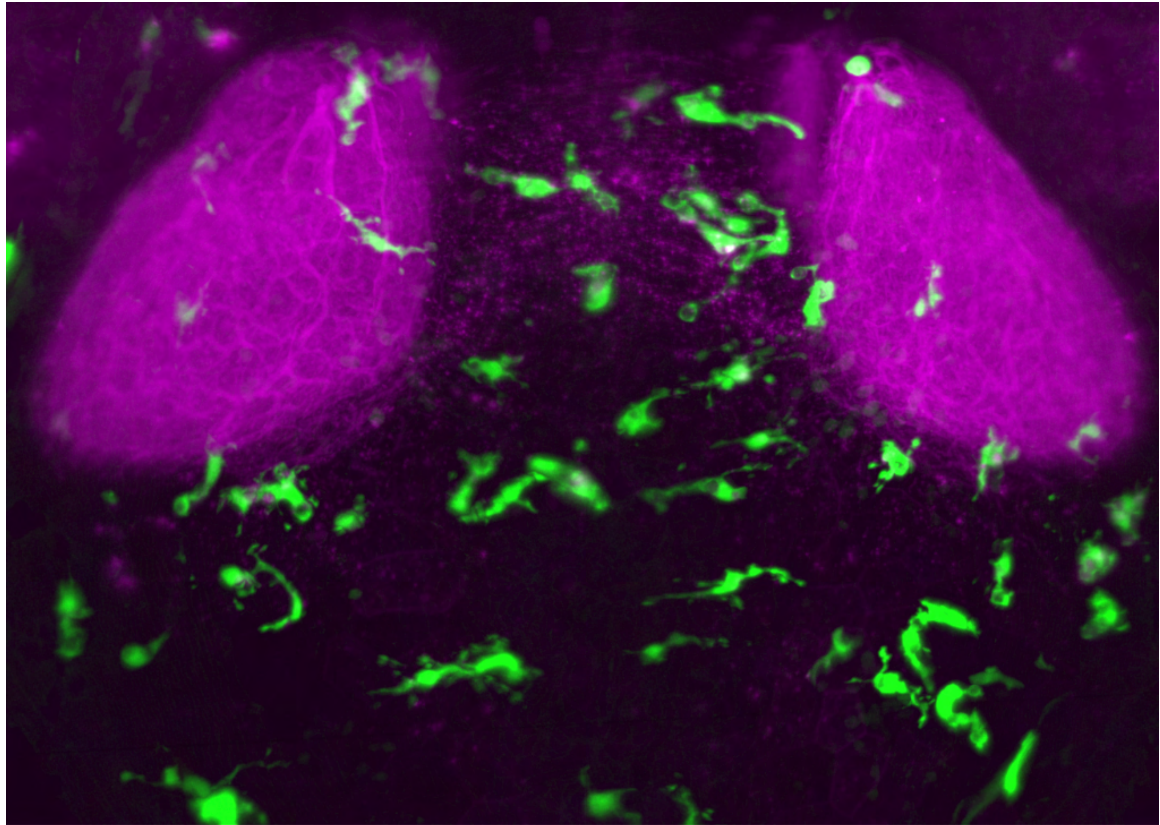
Figure 16: Microglia Apoptosis, Phagocytosis, and Mitosis

Two sets of four still images taken from live-cell lightsheet timelapses of 11 dpf *lakritz* larvae. The cells shown were labeled using Tg(mpeg:GFP). Images A - D showcase the process of one microglia rounding and blebbing before appearing to undergo apoptosis (bottom right cell in A and B), and then being engulfed by another microglia that sensed (B), migrated toward it (C), and then phagocytosed it. Images E - H showcase the process of a microglia entering (E), rounding up (F), and completing mitosis (G-H).

Quantification of Microglia between WT and *lak*

There were not many immediately apparent qualitative differences in microglia dynamics between WT and *lak* (Figure 17; *top & middle*), but I was able to quantify single 5dpf timepoints for both groups. In the individual time points, I found that there were more microglia in the *lak* OT than WT (Figure 17; *bottom left*). I also found that microglia had more volume on average in *lak* compared to WT (Figure 17; *bottom middle*). Lastly, when analysing 4 hour timelapses of these larvae, I found that *lak* mutants had much more variation in their speed

throughout the timelapses (indicating that they start and stop more throughout the video) (Figure 17; *bottom right*).



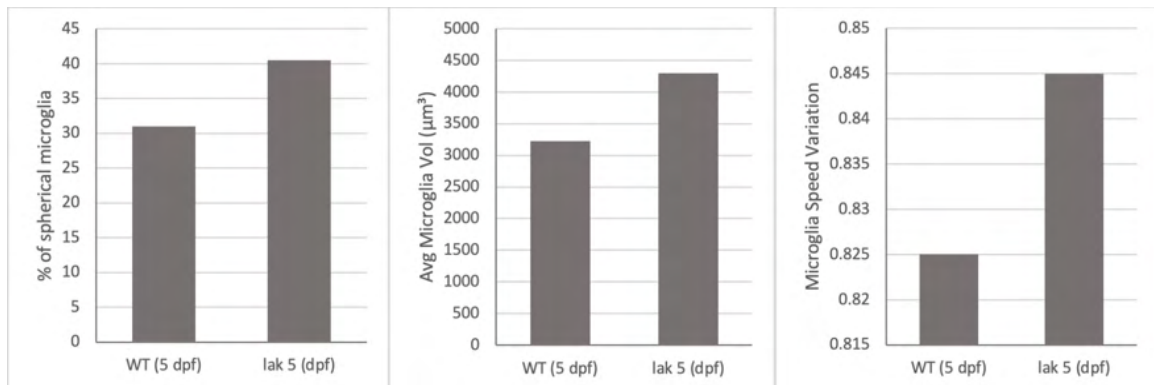


Figure 17: Comparison between WT and *lakritz* microglia at 5 dpf

Two dorsal views of the 5 dpf larval OT. The top image is a WT larva. The middle image is a *lakritz* larva. Magenta is *Tg(ato7:RFP)*, labeling RGCs. Green is *Tg(mpeg1:GFP)*, labeling microglia.

The bottom section is three graphs: the left graph is the percentage of microglia that fall above a certain sphericity threshold (0.695) in WT (n=X) and *lak* (n=Y). The middle graph is the average volume of microglia in the OT. The right graph is the variation in microglia speed (how much the movement speed of the microglia varies throughout the timelapse) in WT and *lak*.

Discussion

This thesis has resulted in over 100 processed images of varying qualities from dozens of different larvae, as well a gradual understanding of the imaging process and how to fine tune and optimize it. Most of these images proved useful to an extent, but many of them also failed in some regards. A few of them provided clear views that were amenable to qualitative and quantitative analyses, showing new cellular dynamics and interactions never before described in zebrafish

RGC growth and OPC migration

A lot of information can be gleaned from the 2-3 dpf timelapse in the WT larva. Beyond just describing the movements and time-course of interactions between RGC axons and OPCs, this video and associated images potentially give preliminary mechanistic insight into these interactions. It is possible that the optic nerve sends important signals to the *nkx2.2a*⁺ cell structures that cue early migration of OPCs into the OT. Specifically, the early RGC projections that end up occupying the more medial neuropil layer (likely the SGC), might trigger migration of *nkx2.2a*⁺ cells, and potentially specification of OPC cells. Soon after the first RGC projections encounter *nkx2.2a*⁺ structures, *nkx2.2a*⁺ cell bodies can be seen migrating up the pre-optic tract (Figure 10), and then later on, cells with OPC morphology migrate alongside the optic nerve to the OT (Figure 10 C & D).

Including qualitative and quantitative analysis of later timepoints indicates that the optic nerve potentially provides a specific migration route for OT development. However, there are a variety of ways OPCs migrate into the OT, especially at 3 dpf (Figure 13), and many OPCs originate from the more medial ventral *nkx2.2a*⁺ region.

RGC associated OPCs

Figures 14 and 15 showcase the overall structure that OPCs form in the OT. WT larvae have a well ordered medial basket structure, as well as more lateral OPCs throughout the neuropil. However, *lak* mutants have a less ordered structure. This potentially indicates that the RGCs regulate the structure of OPCs, either directly through signaling, or indirectly, through other impacts of a missing optic nerve, such as altered OT neuron morphology.

Additionally, *lak* mutants have fewer OPCs overall (Figure 15; *bottom left*). This could indicate that there are specific OPCs associated with RGC axons that differ from the OPCs that make up the basket structure due to the signals they receive. This could be a potential reason for the reduced number of OPCs in *lak* larvae. A hypothesis based around this idea is that the optic nerve sends signals to *nkx2.2a*⁺ cells that encourage them to differentiate into OPCs that are specifically associated with RGC axons. Another hypothesis is that the optic nerve might provide a migration route for OPCs, and without it, there are fewer OPCs able to migrate to the OT. One could try to characterize these migration differences by quantifying differences in the number of OPCs that migrate into the OT from different regions between WT and *lak* in early (2-4 dpf) compared to later (5-9 dpf) timepoints.

Lastly, the fact that OPC number seems to increase from 5-9 dpf in *lak* mutants, could demonstrate a different mechanism for maintaining an ideal OPC population. It is possible that fewer OPCs migrating to the OT initially could be made up for by additional migration later on, or increased proliferation of OPCs already in the OT. A potential avenue of future study in the lab could be to try to determine if there is a quantitative difference in OPC proliferation or migration in *lak* mutants. The OPC timelapses I used for this preliminary study could potentially be analyzed further, along with additional samples to try to answer this question.

Questions from *nkx2.2a* labeling specificity

The basic characterization, provided in the results section, of cell morphologies and structures labeled by the *Tg[nkx2.2a:GFP]* reporter was mainly just for a better understanding of how the transgenic marker works, and what it labels. However, It raises some additional questions. Specifically, the cells with radial-glia-like morphology are very interesting. Radial glia cells are expected to exist along the pial surface of the OT, however, *the nkx2.2a:GFP* + cells with long intertectal projections exist adjacent to the neuropil, and then extend up to occupy the very dorsal and anterior surface of the OT. It is unclear what these cells are, but there seemed to be an increased amount in the 5-7 dpf *lak* larvae, whereas in WT larvae, these cells were very dynamic and complex at 3 dpf, but seemed to decrease in movement and amount by 5 dpf (or they lost *nkx2.2a* expression). If these cells are functionally similar to radial glia cells, or if they are in-fact radial glia cells, then maybe there are some differences in radial glia cells and how they function in regards to OPCs in *lak* mutants compared to WT.

Microglia

The quantitative analysis of the microglia images and timelapses was very preliminary, only using two larvae. I am not confident in the robustness of these results, but they do demonstrate a potential trend in how microglia differ between WT and *lakritz* (Figure 17).

The observed apoptosis, phagocytosis, and mitosis dynamics in Figure 16, were exciting because they demonstrated that these dynamics can be very clearly observed. These observations also indicate the time that these dynamics take to occur. And the steps we would expect to see. Both processes occur in less than an hour. The mitotic microglia in Figure 17 D-H gives an example of how microglia replenish their numbers in the OT. It's unclear why this microglia underwent mitosis, but there are likely competitive mechanisms at play determining the

localization and spacing of microglia, and it may have received a signal indicating a need for more microglia.

The phagocytic microglia in Figure 16 A-E, reacts very quickly to the apoptotic microglia, and likely reacts based on secreted signals from the apoptotic cell, causing it to first send exploratory projections in the direction of the cell, and then it retracts them and send a very quick and direct projection to start the engulfment process. This helps us to understand how quickly microglia react to apoptotic cells in the area. It also indicates that, when studying cell death in the OT, it is possible that apoptotic cells are cleared very rapidly, which would contribute to variation among organisms and make comparison of cell death between organisms difficult.

Future directions

There are many different directions someone could go based on this study. First of all, there was a very low sample size for every group, so any results should be able to be replicated in additional WT and *lak* larvae. I got particularly few viable *mpeg:GFP / atoh7:RFP* timelapses, so there could be more timelapses taken for both WT and *lak* from 5-9 dpf.

Additionally, one of the most powerful timelapses I got was in the 2-3 dpf WT larva labeled with *nkx2.2a:GFP* and *atoh7:RFP*, but I wasn't able to get a *lak* timelapse at the same time point.

I took many images and timelapses, and while I looked through all of the robust ones fairly thoroughly, there are likely differences in patterns that are visible in OPC and microglia dynamics.

The current *nkx2.2a:GFP* reporter results in all OT OPCs being labeled, but this means that the individual morphologies and arborization patterns of OPCs cannot be characterized. Developing a method to sparsely label OPCs, such as with a plasmid injection, would allow one to study individual OPC dynamics better.

There is also the potential of looking closer at *nkx2.2*+ radial glia-like cells, and comparing with labeled radial glia cells, and then determining if there are differences in radial glia behaviors between WT and *lak*.

I had intended to look at calcium signaling dynamics in the OT between WT and *lak*, but didn't get a chance to. I tried getting a heat-shock GCaMP plasmid to work, but only got sparse static labeling of neurons, and it's likely this was in-part due to the fact that our anesthetic (tricaine methanesulfonate) blocks sodium ions from entering the cell and thus blocks action potentials. We bought a different anesthetic (D-tubocurarine chloride pentahydrate), but never got a chance to test it. This experiment would require genotyping adult (GCaMP), then pairing them with *lak* +/- heterozygote adults, waiting for them to mature, then genotyping those adults, and then pairing them and imaging both the *lak* -/- and +/- (WT) larvae and observing the calcium activity of OT cells.

Lastly, we know almost nothing about the structure of the OT at later time points in *lakritz* mutants, and how it is integrated with other neural circuitry. The TSc is a brain structure ventral and lateral to the OT, and it is involved in processing of sensory information from the zebrafish lateral line. It could be very interesting to see how the *lakritz* OT functions and is structured when it develops without the presence of the optic nerve, especially in regards to connections to the TSc and other brain structures.

Appendix A: Additional Protocols

Embryo Micro-Injections

Once the eggs from the paired fish were laid and fertilized, they were collected by straining the breeding box water. Then they were lined up along the edge of a glass slide to be injected. To set up the micro-injection rig, a glass capillary tube was melted and pulled into a fine needle point and then filled with a few μm of injection solution consisting of Annexin V:GFP (van Ham et al., 2010) plasmid containing TOL2 transposon integration sites, and Tol2 transposase mRNA. in nano-pure H_2O . The needle was then affixed to the rig, and the very tip was broken off such that the correct volume of solution (1-2 nl) could be injected with each puff of pressure from the rig. The needle would then be lined up and manipulated to puncture through the embryo chorion and single cell membrane to inject into the cytoplasm or alternatively into the yolk on the border with the single cell.

Many of the embryos by the end of the injections had undergone their first or second division cycles and had two or four cells. These cases could still be injected by placing the needle tip in the yolk sac at the border near where the cells meet.

Enucleations

At five dpf, ~10-15 larvae were selected for the experimental enucleated group and ~10-15 larvae were selected for the control WT group. The

experimental group was separated, and then anesthetized by adding enough 20x tricaine mesylate to achieve a 1x concentration in the housing dish (0.0015% w/v). Once the larvae were anesthetized, they were transferred one by one to a dry small dish lid. A large drop of 1% low melting point agarose was added on top of each larva, and they were oriented such that their left eye was facing upwards towards the microscope objective. Once 5-6 larvae had been mounted and the agarose had set for all of them, a large drop of tricaineated water was added to cover the agarose and keep it moistened and to keep the larvae anesthetized.

A tungsten needle was sharpened to a very fine straight point using electrolysis. Then, the needle was used to separate the eye from the surrounding tissue by inserting it at the ventral edge of the eye and carefully sliding around the perimeter of the eye making sure to keep the needle very shallow so as to not damage any deeper tissue or brain. Once the membrane holding the eye in place was separated, the eye was moved up and to the side in order to access the optic nerve and the deeper tissues connected to the eye. A slightly more hooked needle was used to gently tear at the nerve without breaking open the eye or damaging the remaining tissue too much. Once the eye was completely severed from the larva, I would move on to the next mounted larva. Once all mounted larvae were enucleated, I would dismount them by cutting the agarose along the length of the body with forceps until the larvae were no longer constrained. This was repeated for one or two more sets of larvae until the whole experimental group was enucleated.

After the surgery, larvae were transferred to a recovery solution consisting of Mark's Modified Ringers solution (MMR), Penicillin-Streptomycin antibiotic, and E3. The control group was simply transferred to the recovery solution around the same time as the experimental.

Appendix B: Additional Figures

Microglia show dynamic interactions with cells undergoing apoptosis

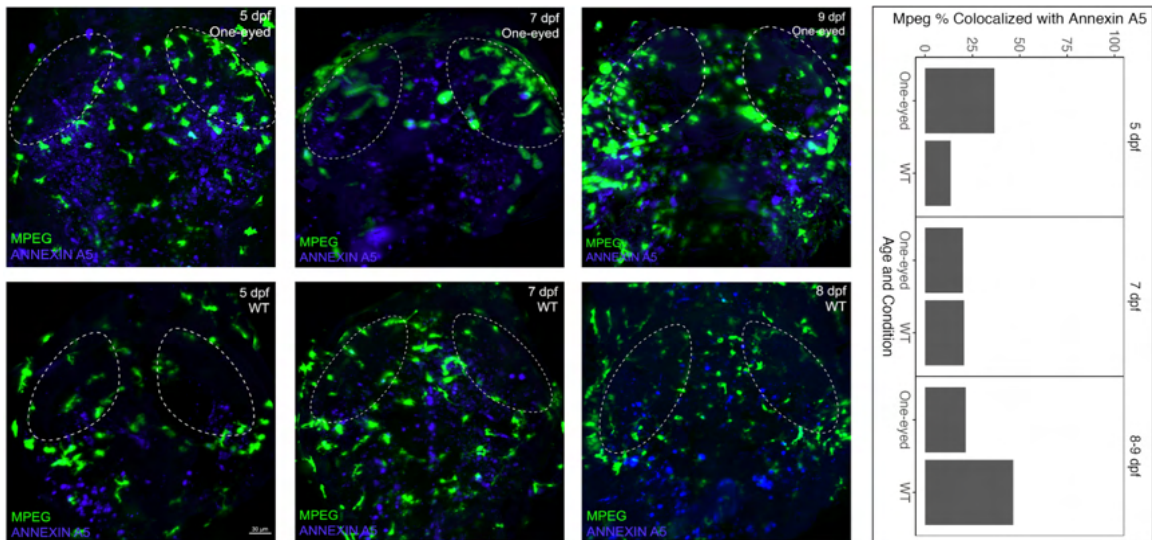


Figure 18: Microglia colocalization with apoptotic cells

Microglia are labeled with Tg(mpeg:mCherry), and apoptotic cells are labeled with Tg(AnnexinV:GFP). Graph shows the percentage of microglia that are colocalized with Annexin A5 + material, indicating microglia phagocytosis of apoptotic cells.

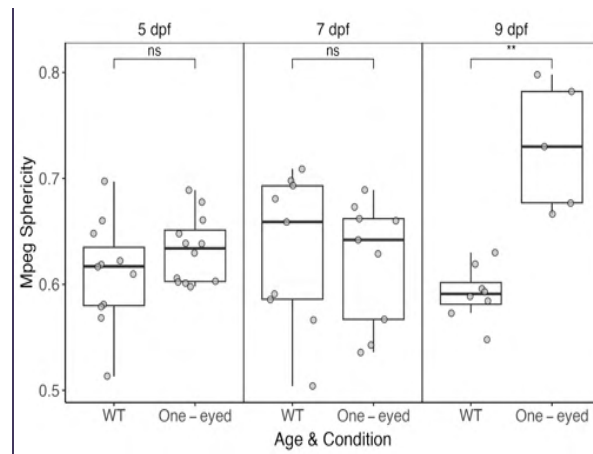
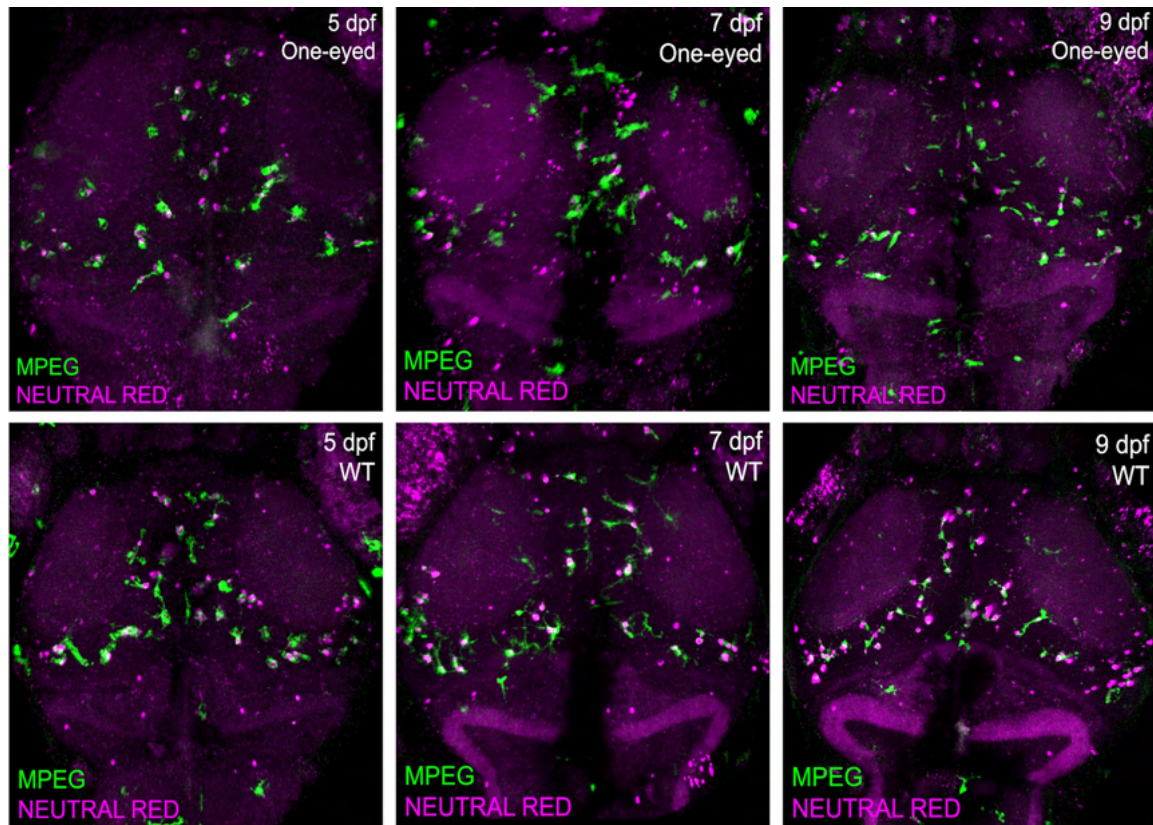


Figure 19: Microglia colocalization with Neutral Red stained lysosomes

Microglia are labeled with Tg(mpeg:GFP) and Neutral Red (NR) was used to stain phagosomes. Graph shows the average sphericity of microglia between WT and One-eyed larvae, at 5, 7 and 9 dpf. Images acquired by Jiahe Yue

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